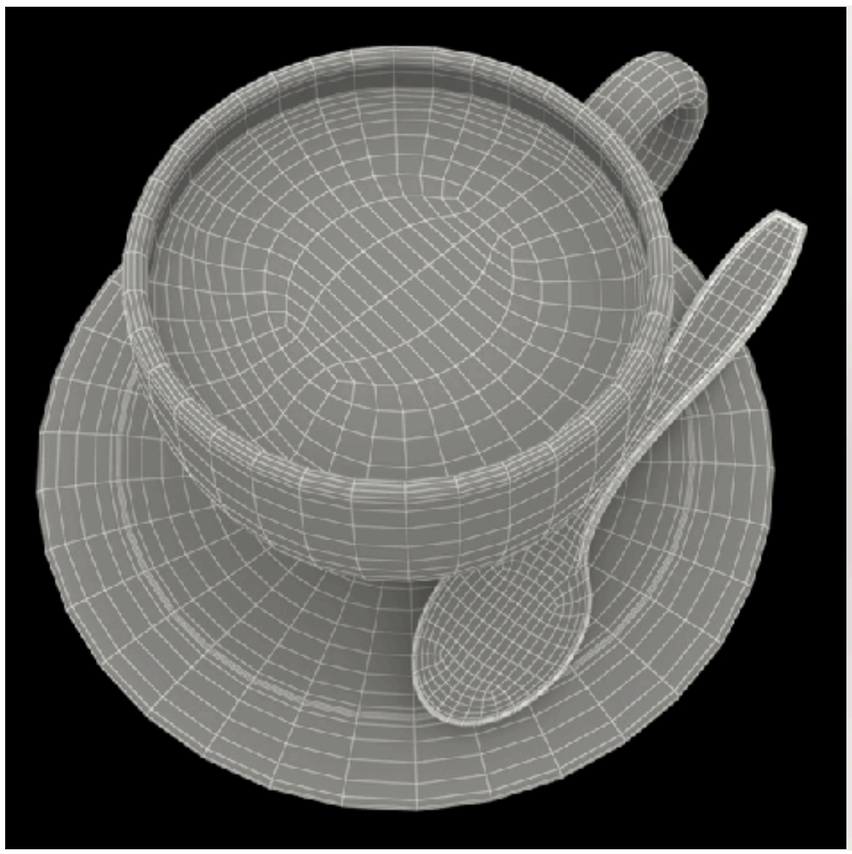
Lecture 14:

Introduction to Material Modeling

Computer Graphics and Imaging
UC Berkeley CS184

What is Material in Computer Graphics?



3D coffee mug model

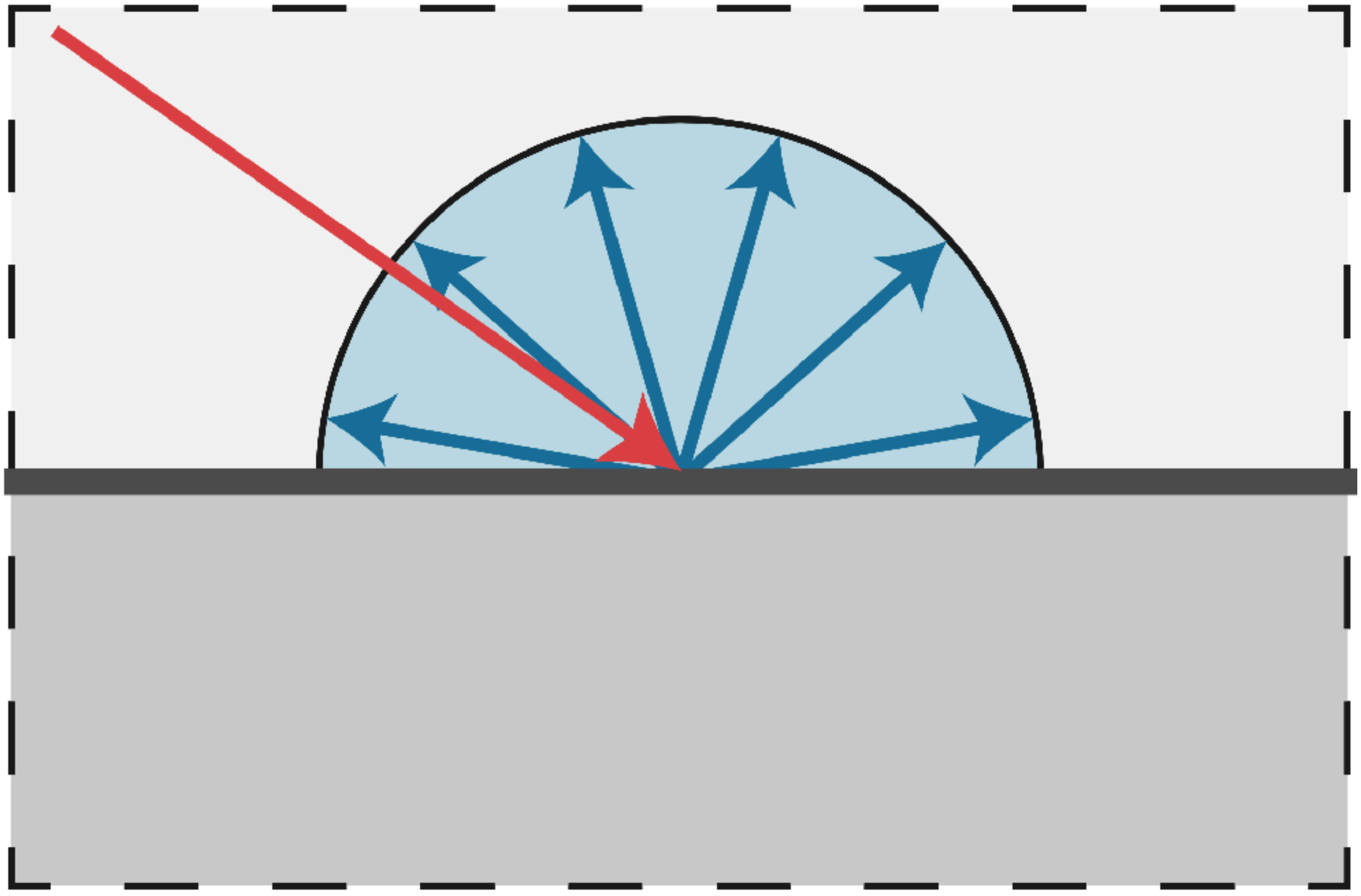
Rendered

Rendered

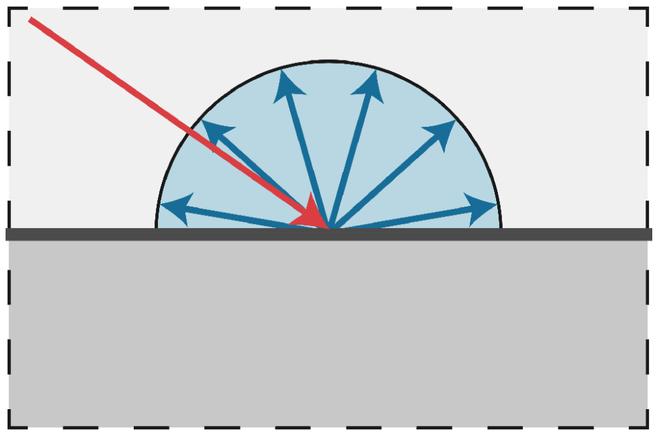
[From TurboSquid, created by artist 3dror]

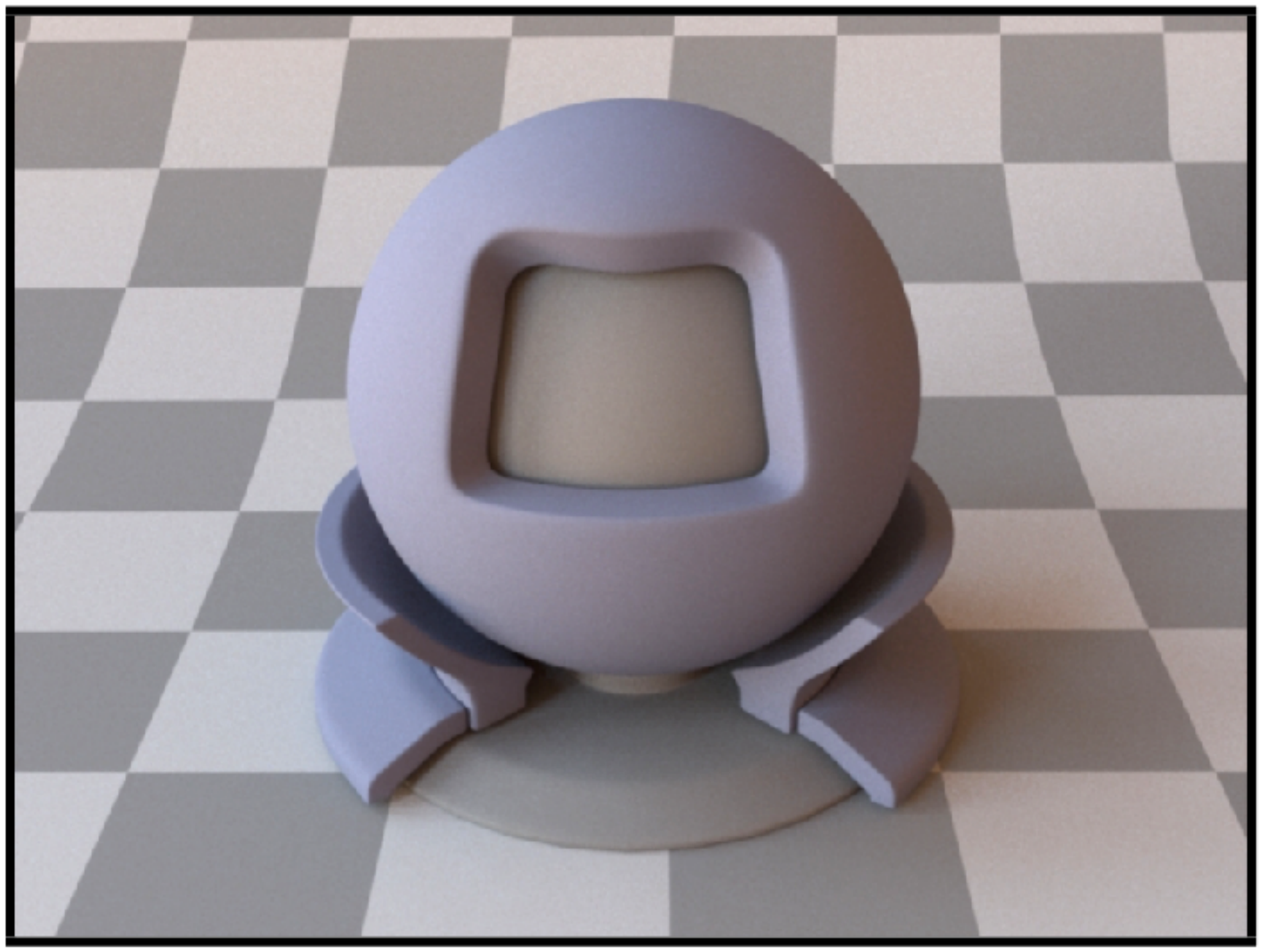
Material == BRDF

What is this material?

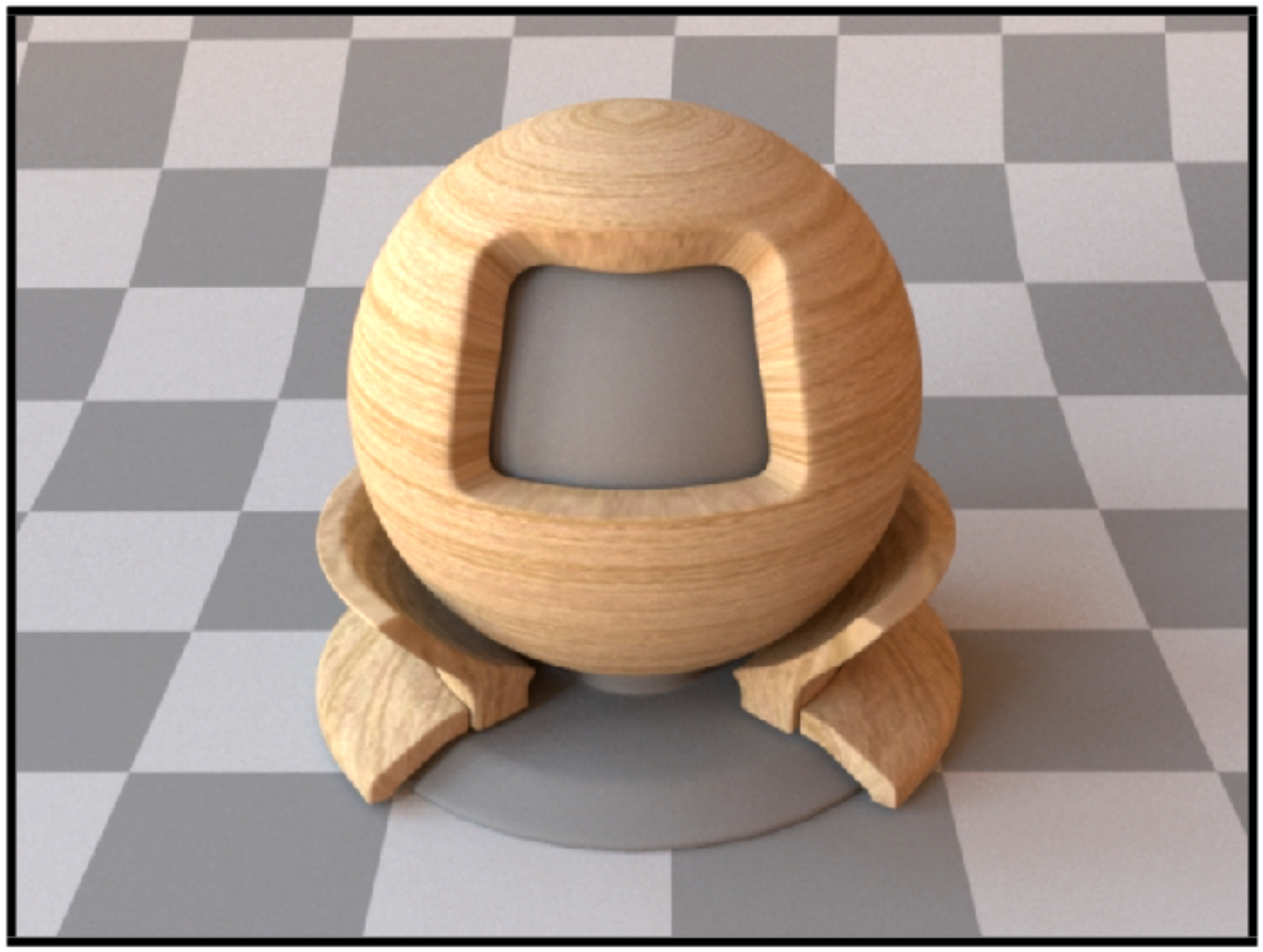


Diffuse / Lambertian Material (BRDF)





Uniform colored diffuse BRDF

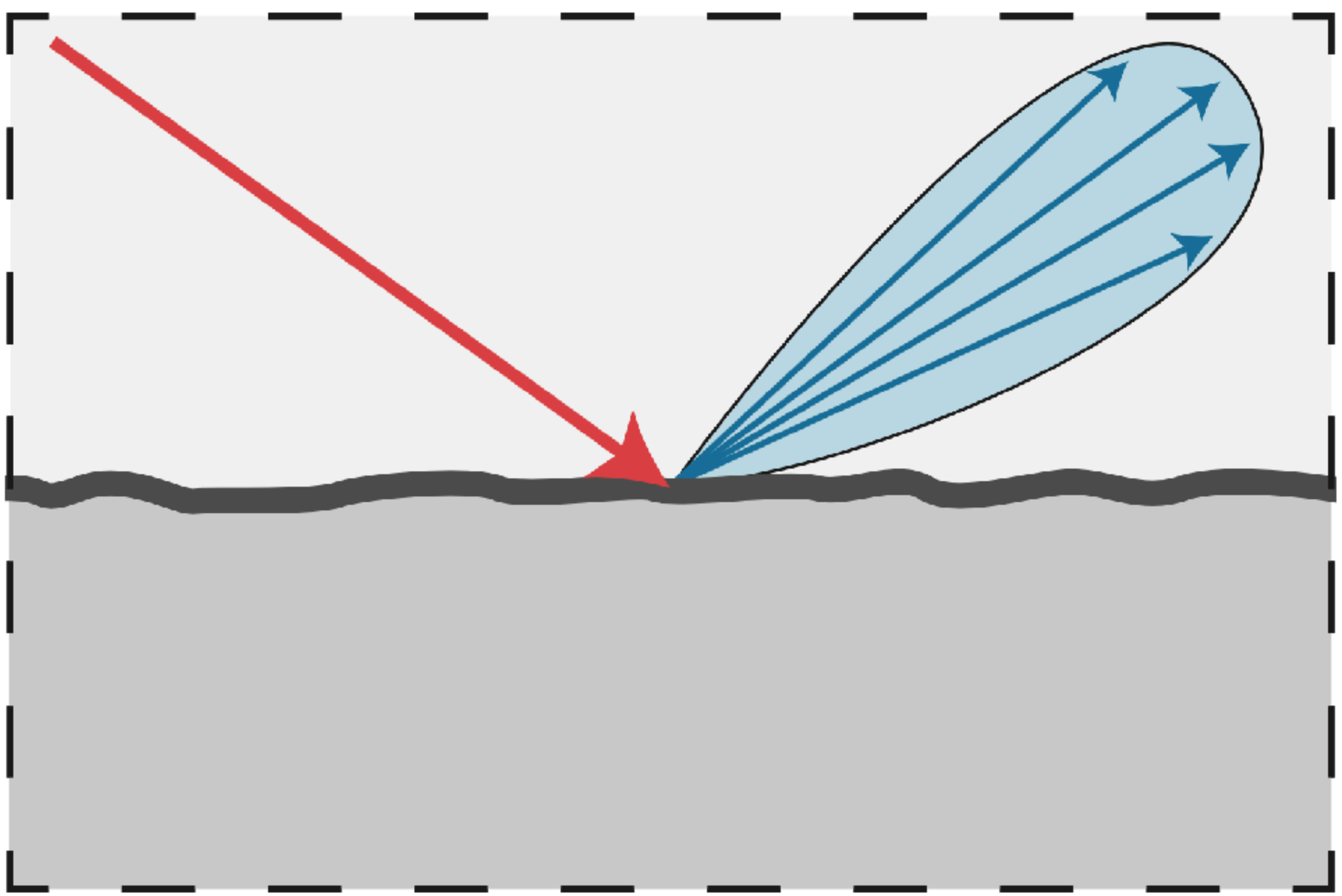


Textured diffuse BRDF

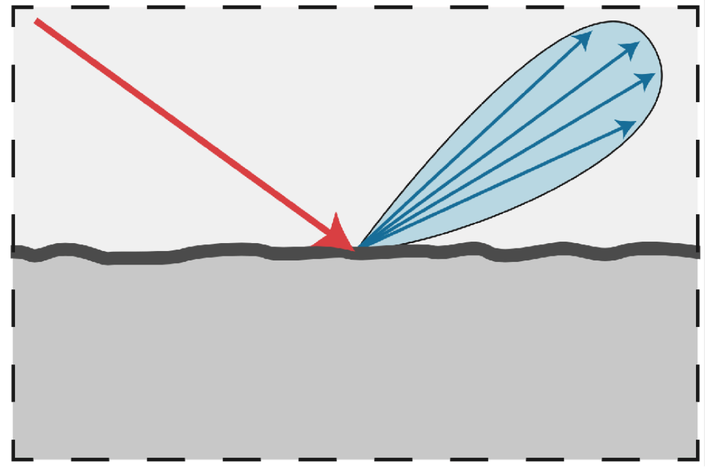
CS184 [Mitsuba renderer, Wenzel Jakob, 2010] Ren Ng

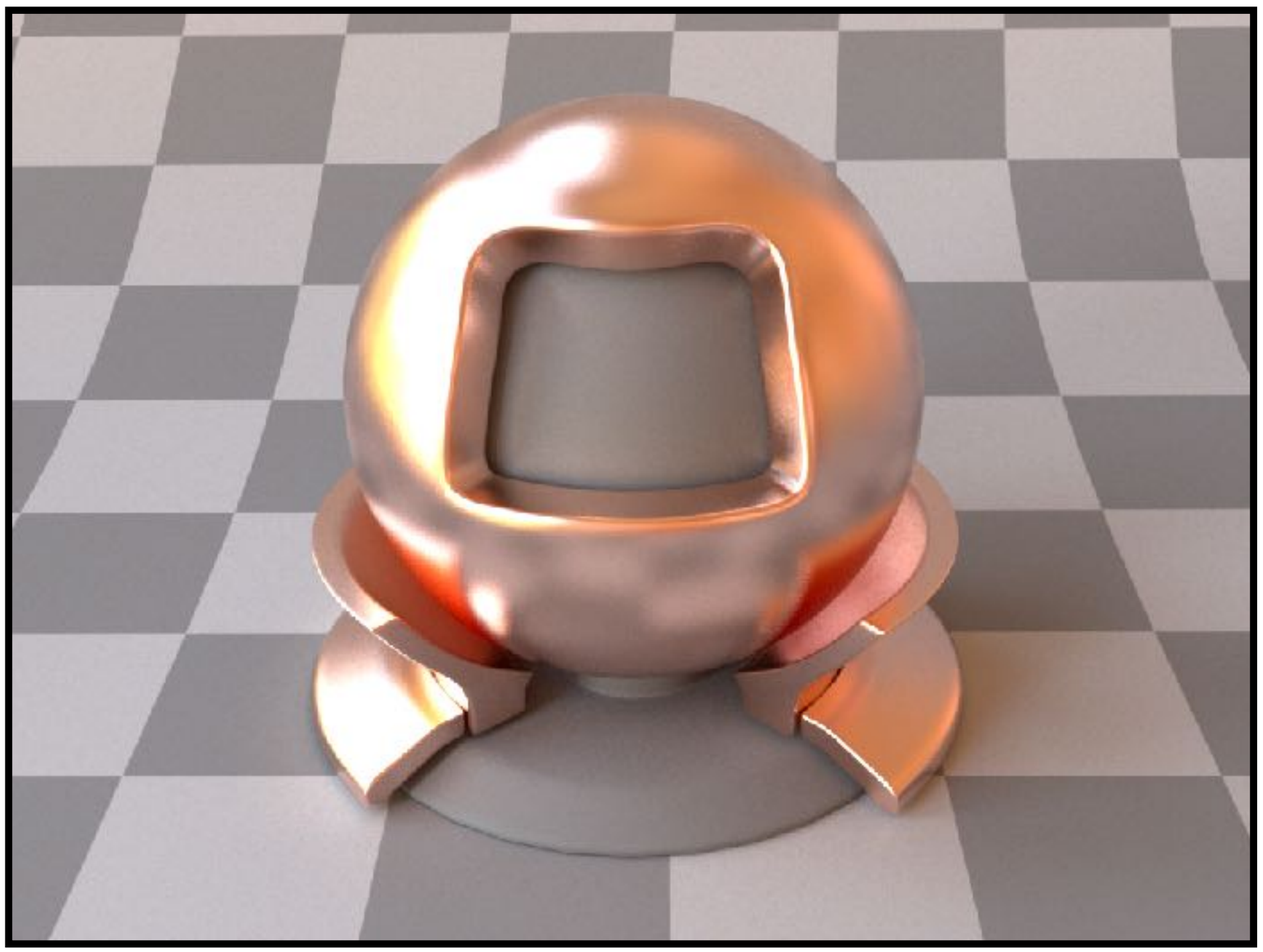
Presented by David McAllister

What is this material?

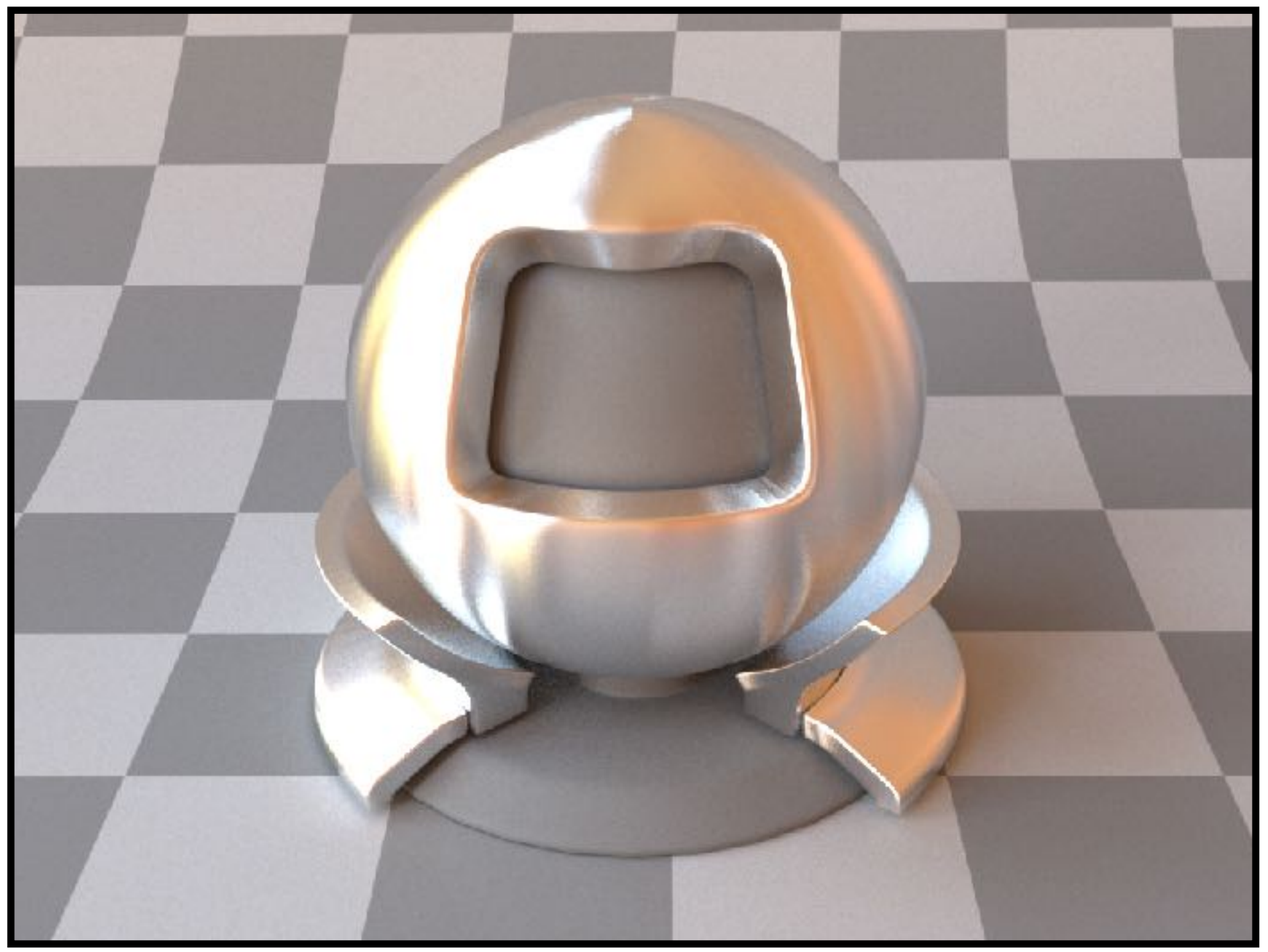


Glossy material (BRDF)



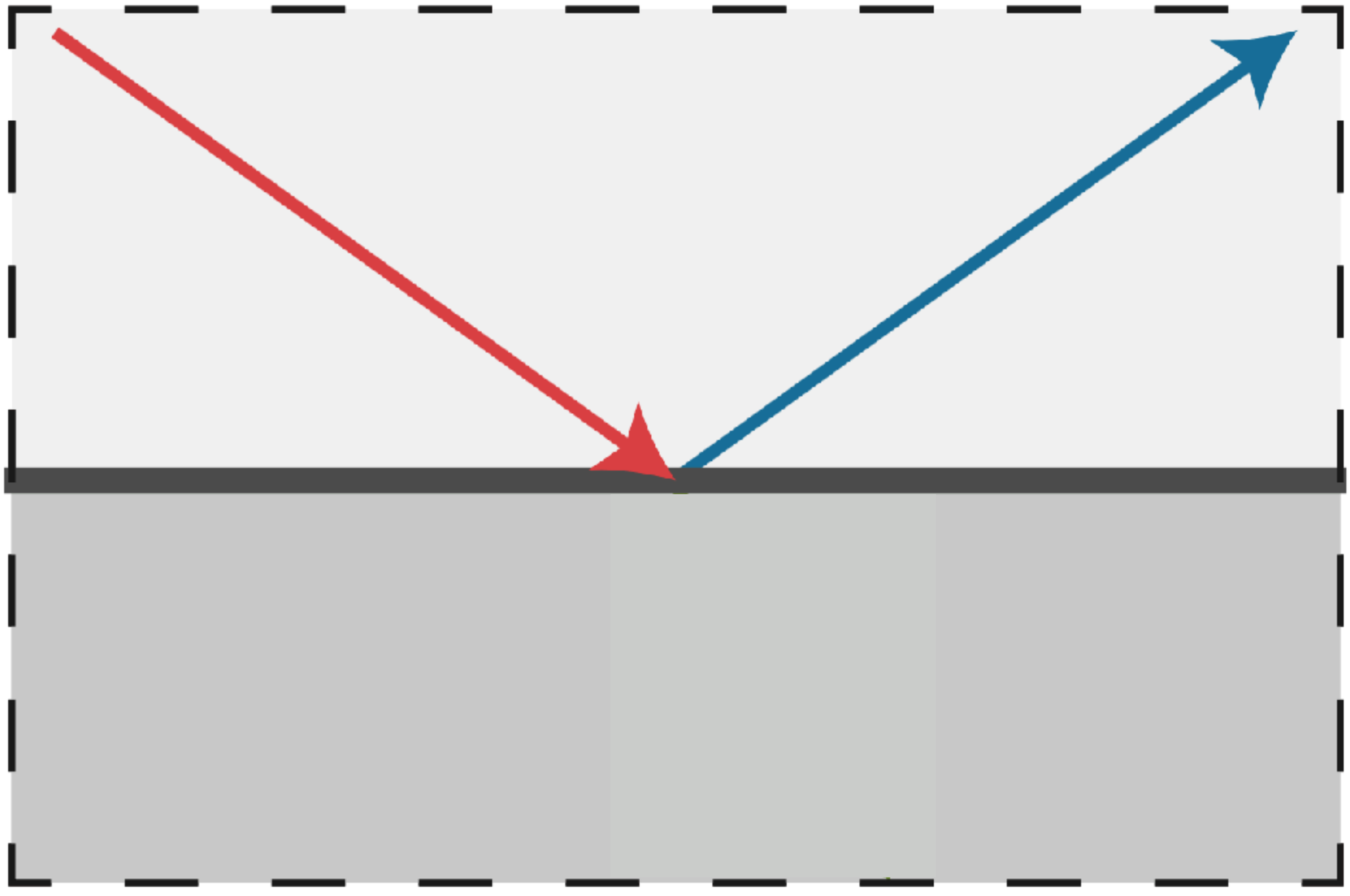


Rough Copper

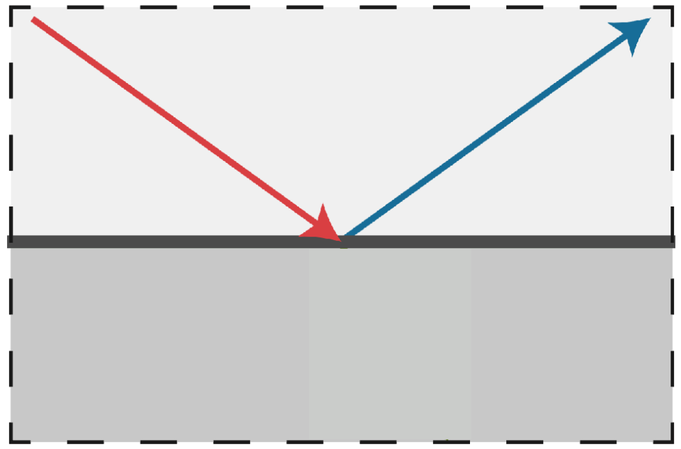


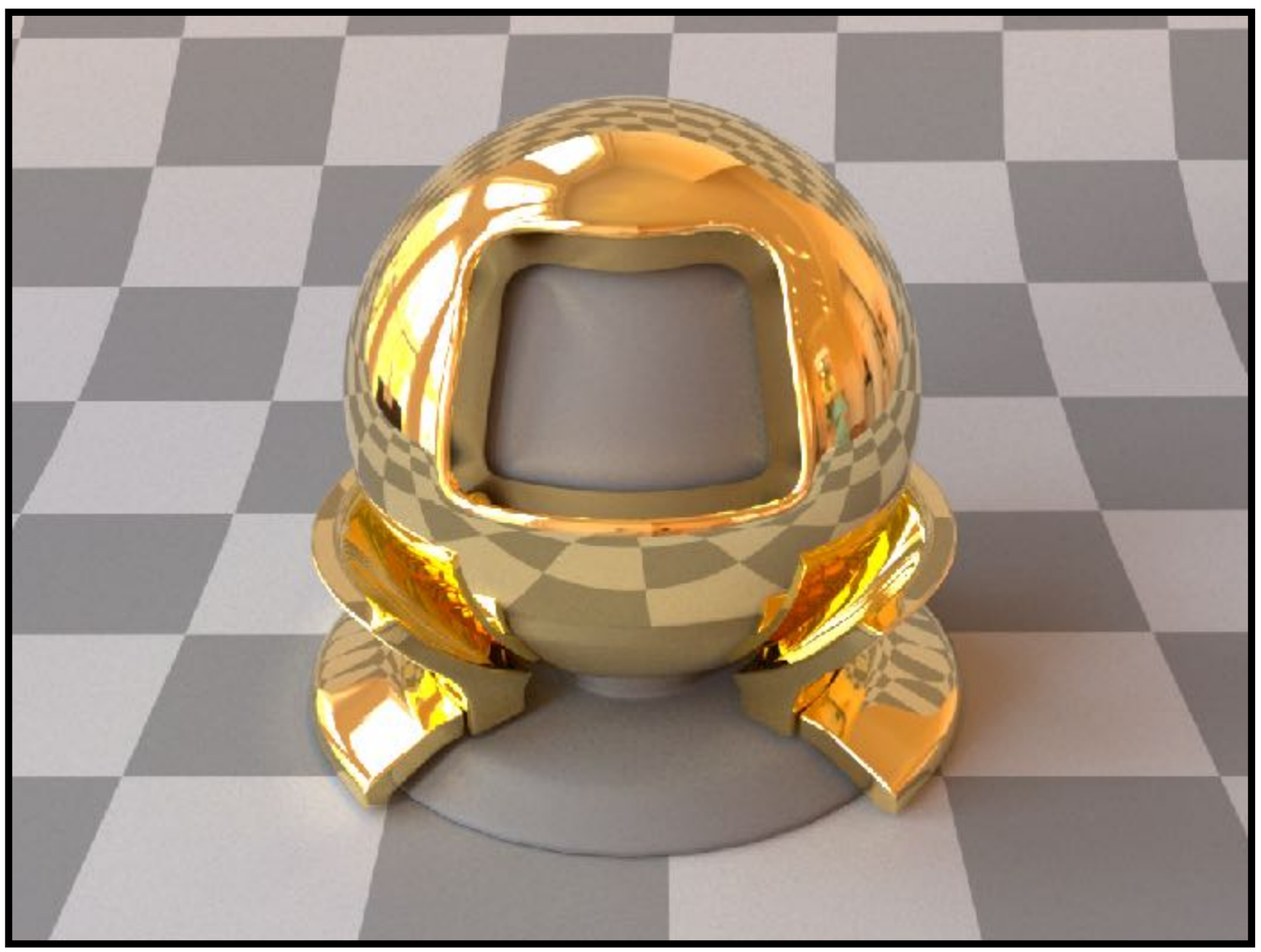
Brushed Aluminum

What is this material?

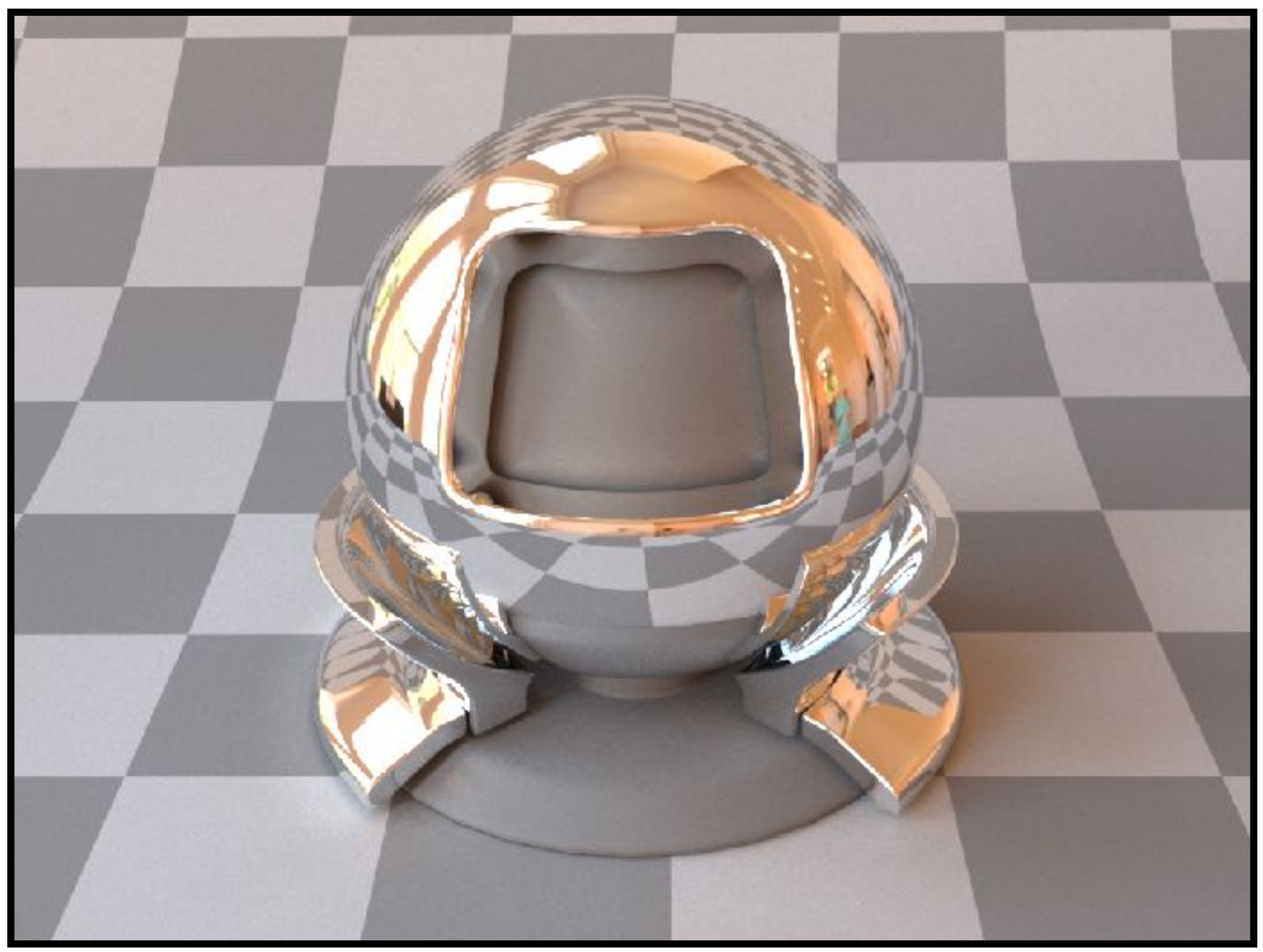


Ideal reflective material (BRDF)



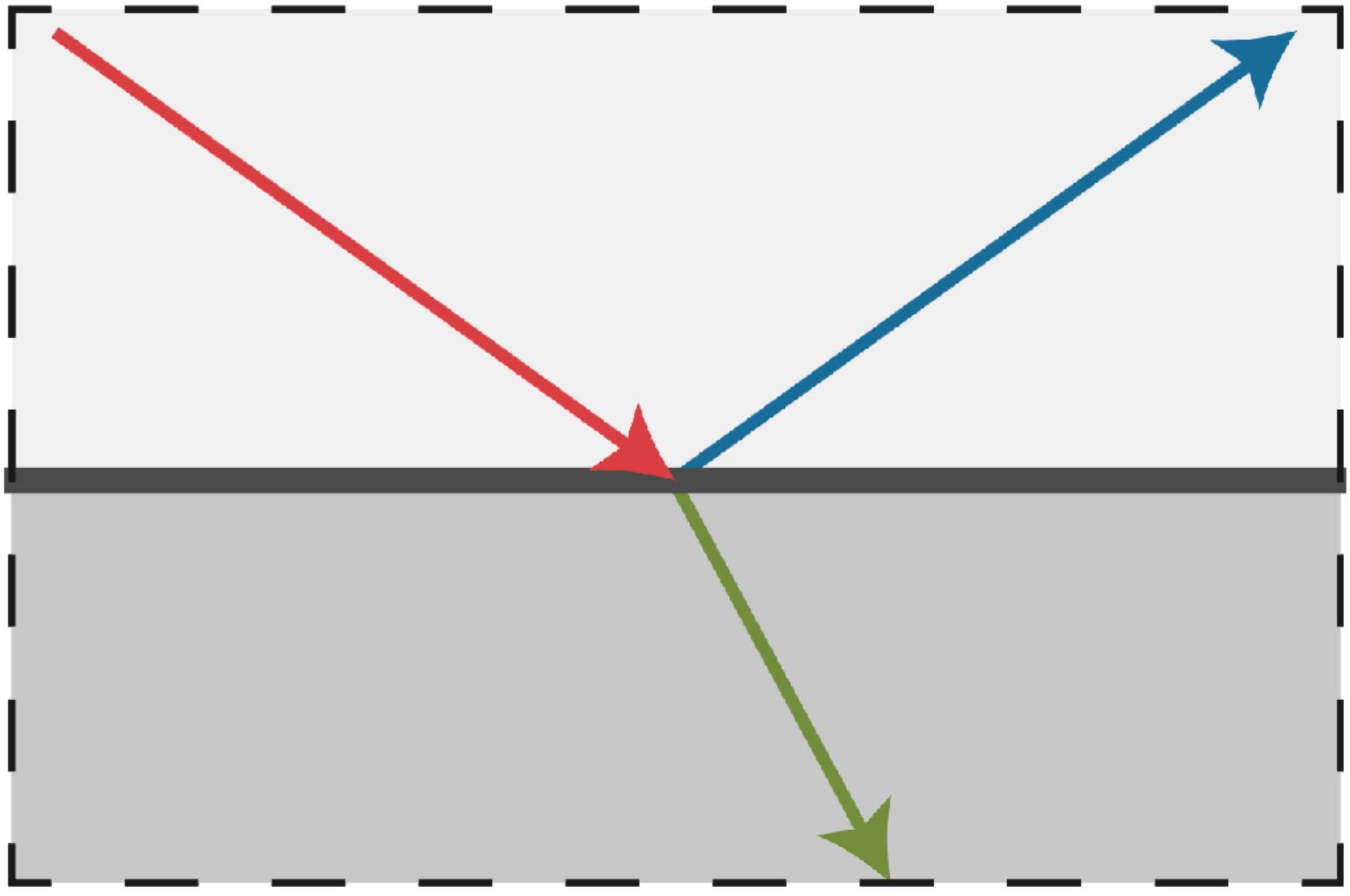


Gold

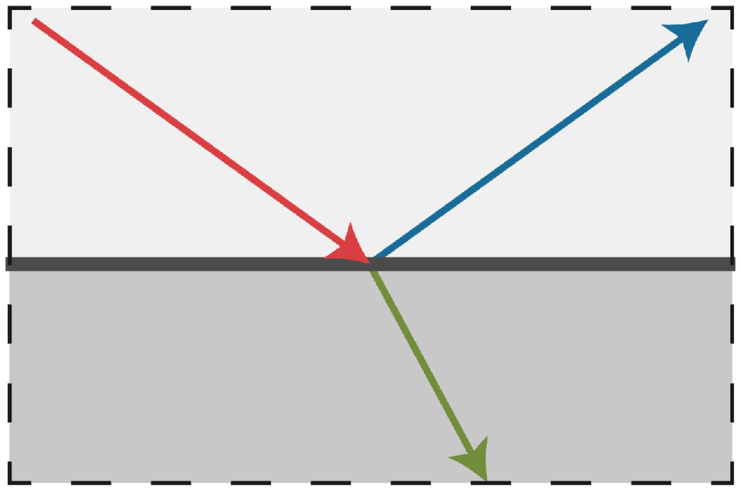


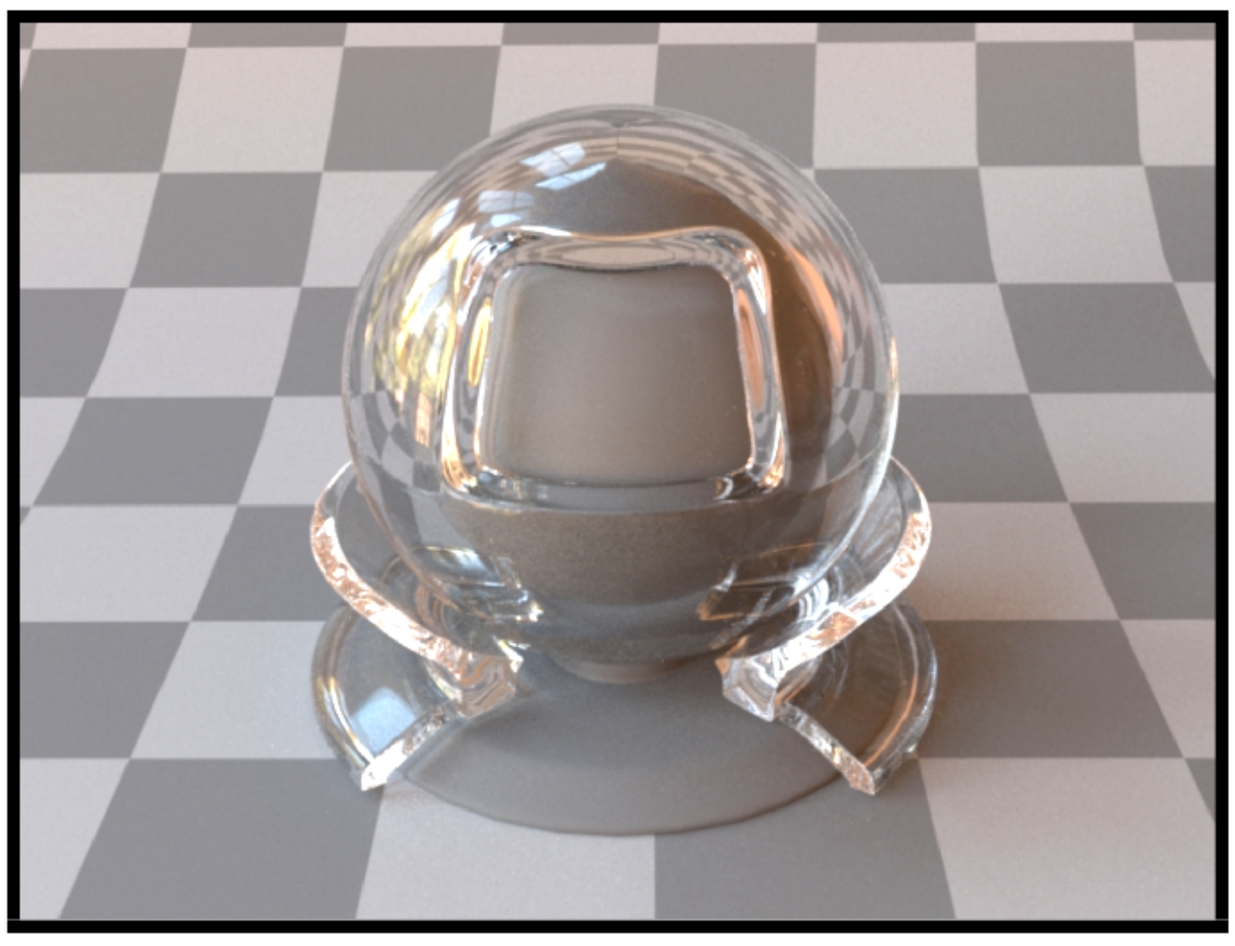
Aluminum

What is this material?

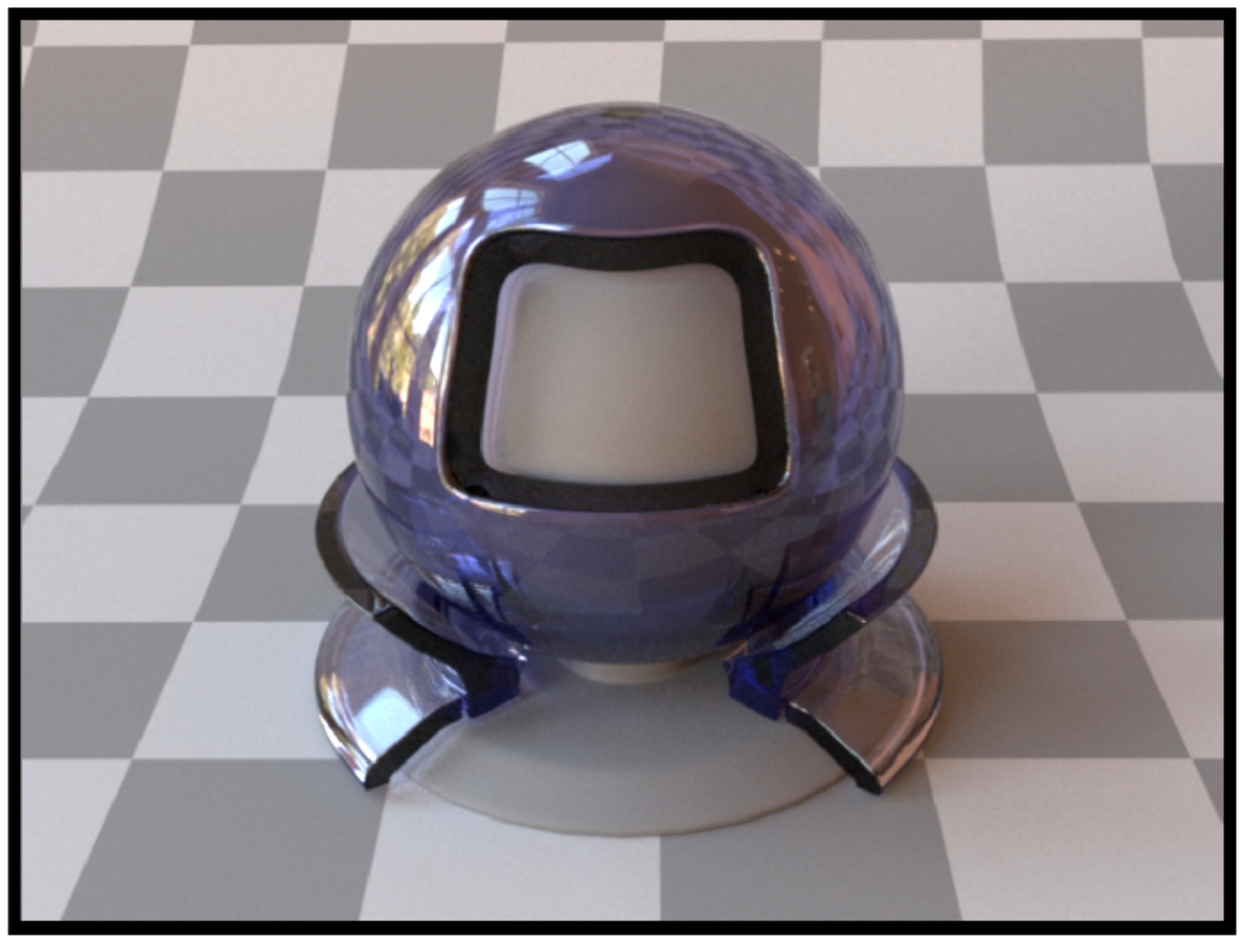


Ideal reflective / refractive material (BSDF*)





Air ←→ plastic interface



Air ←→ glass interface
(with absorption)

CS184 [Mitsuba renderer, Wenzel Jakob, 2010] Ren Ng

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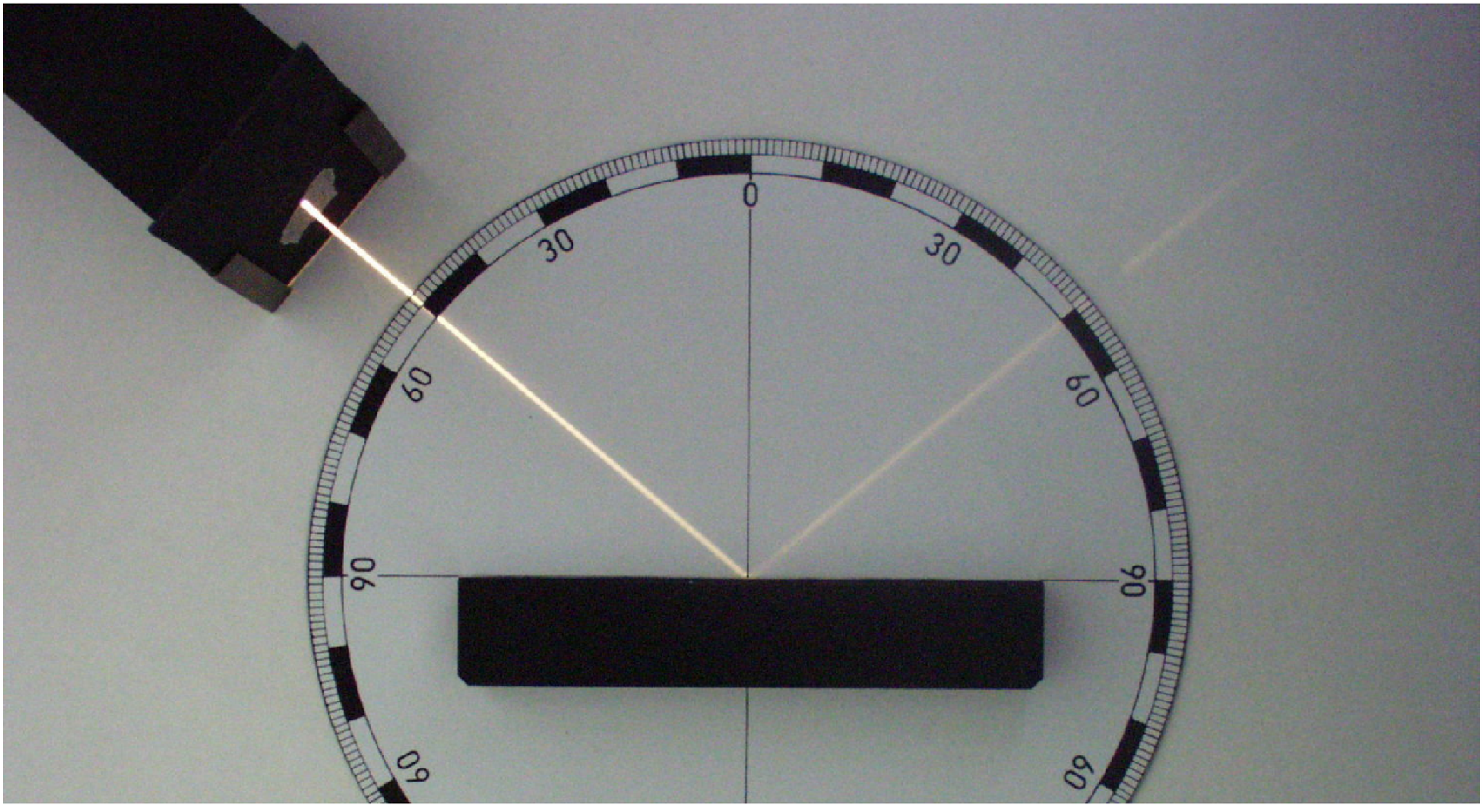
Ideal Reflection and Refraction

Perfect Specular Reflection



[Zátonyi Sándor]

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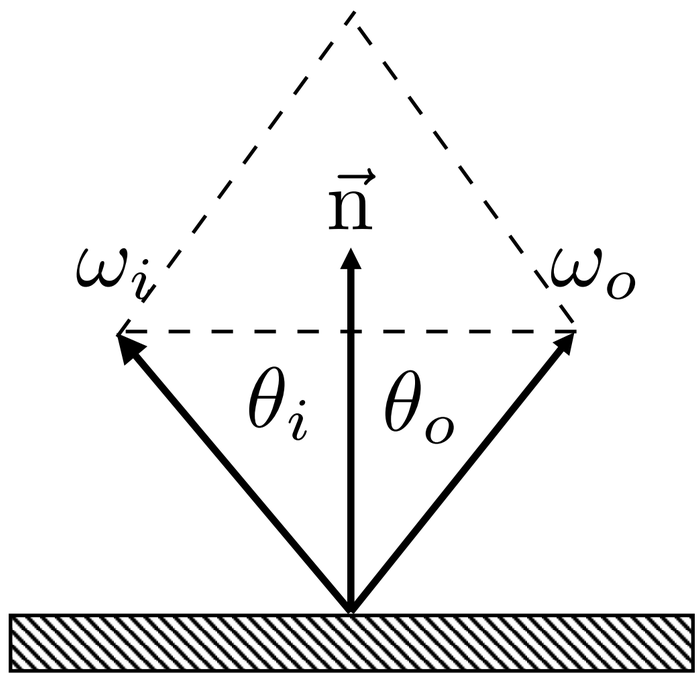
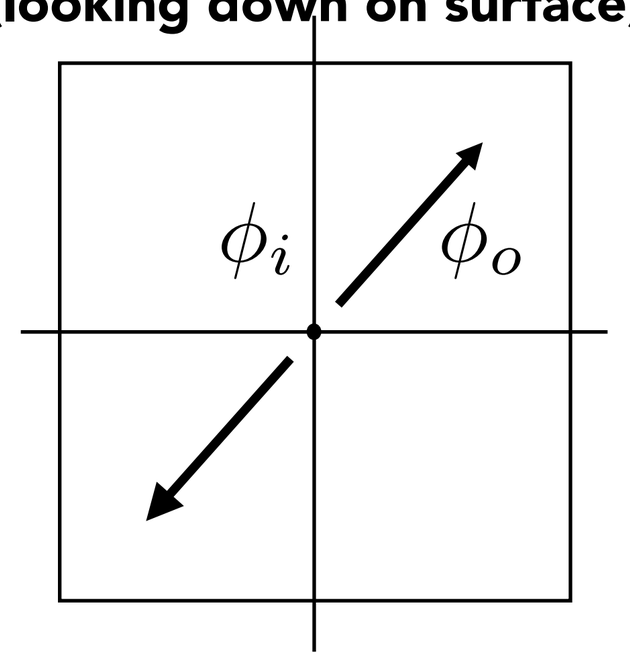
Perfect Specular Reflection



$$\theta = \theta_o = \theta_i$$

Top-down view
(looking down on surface)

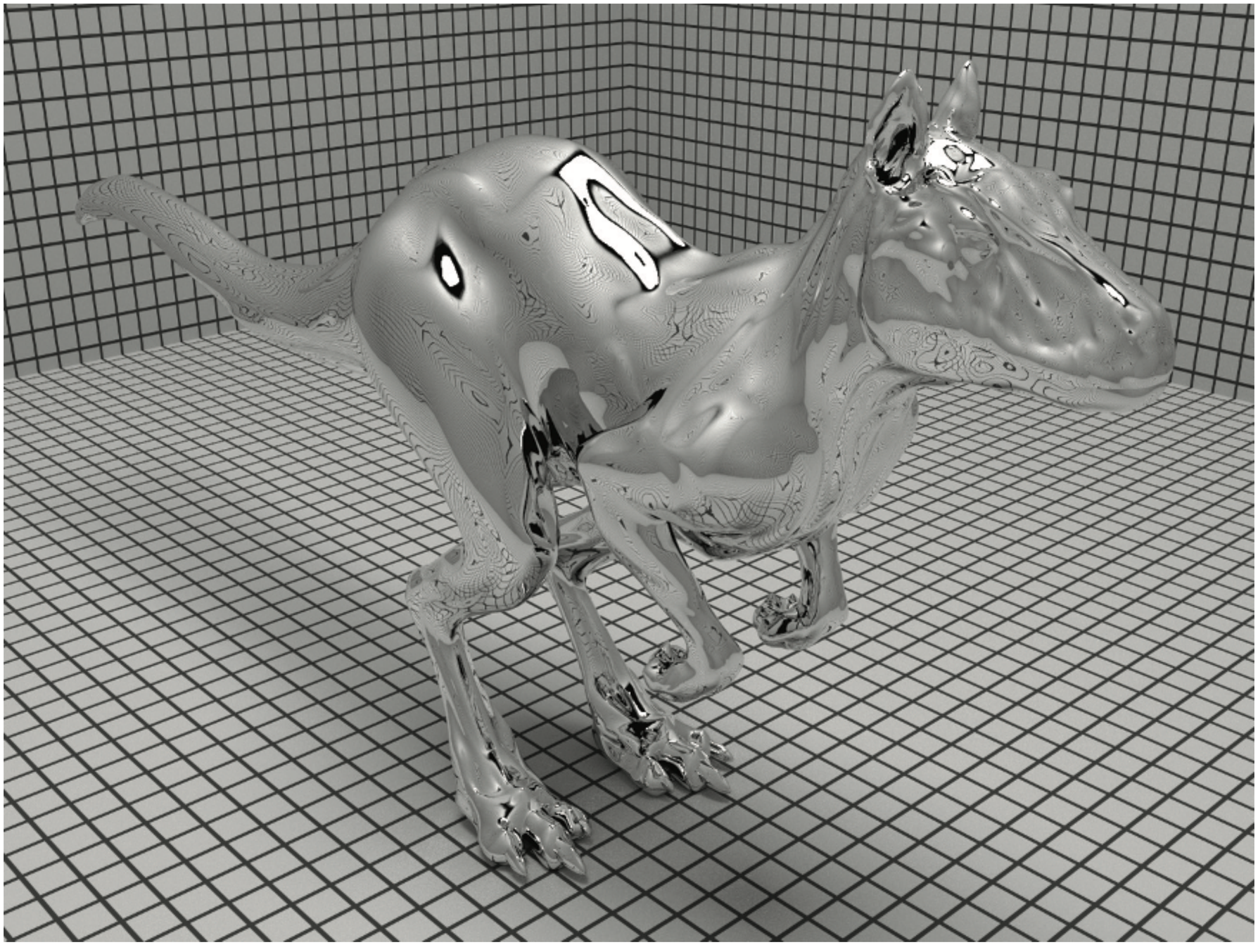


$$\phi_o = (\phi_i + \pi) \bmod 2\pi$$

$$\omega_o + \omega_i = 2\cos\theta\,\vec{n} = 2(\omega_i \cdot \vec{n})\vec{n}$$

$$\omega_o = -\omega_i + 2(\omega_i \cdot \vec{n})\vec{n}$$

Perfect Specular Reflection BRDF



PBRT

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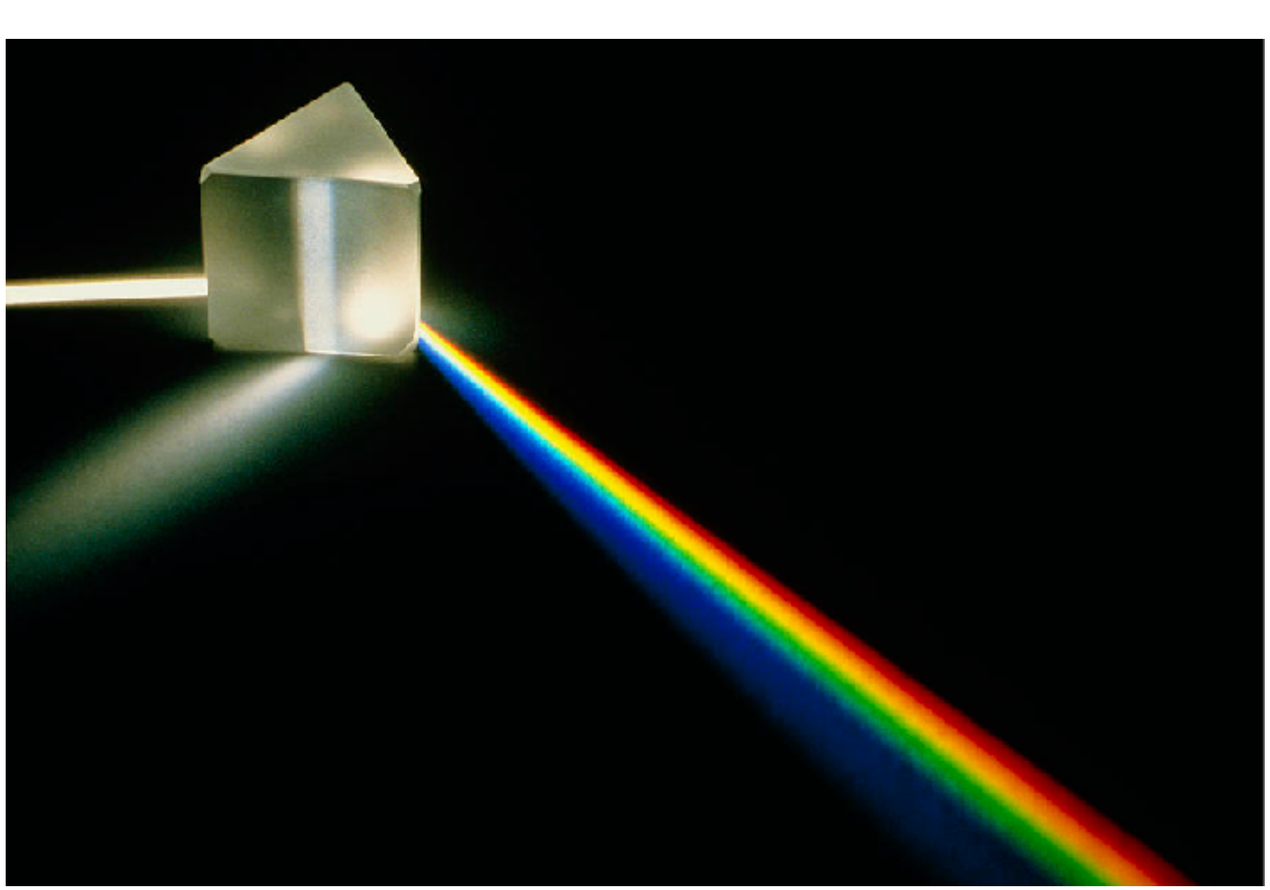
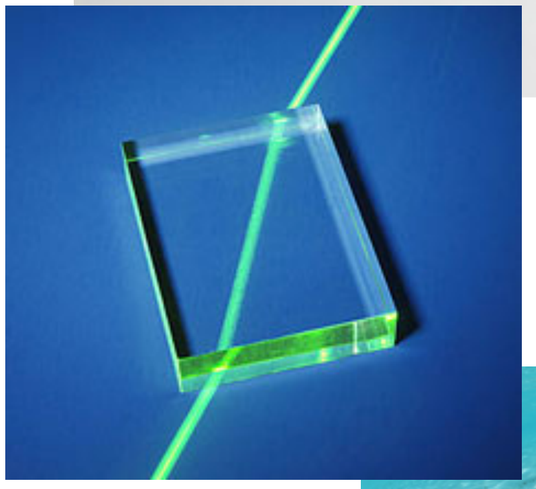
Presented by David McAllister

Specular Refraction

In addition to reflecting off surface, light may be transmitted through surface.

Light refracts when it enters a new medium.





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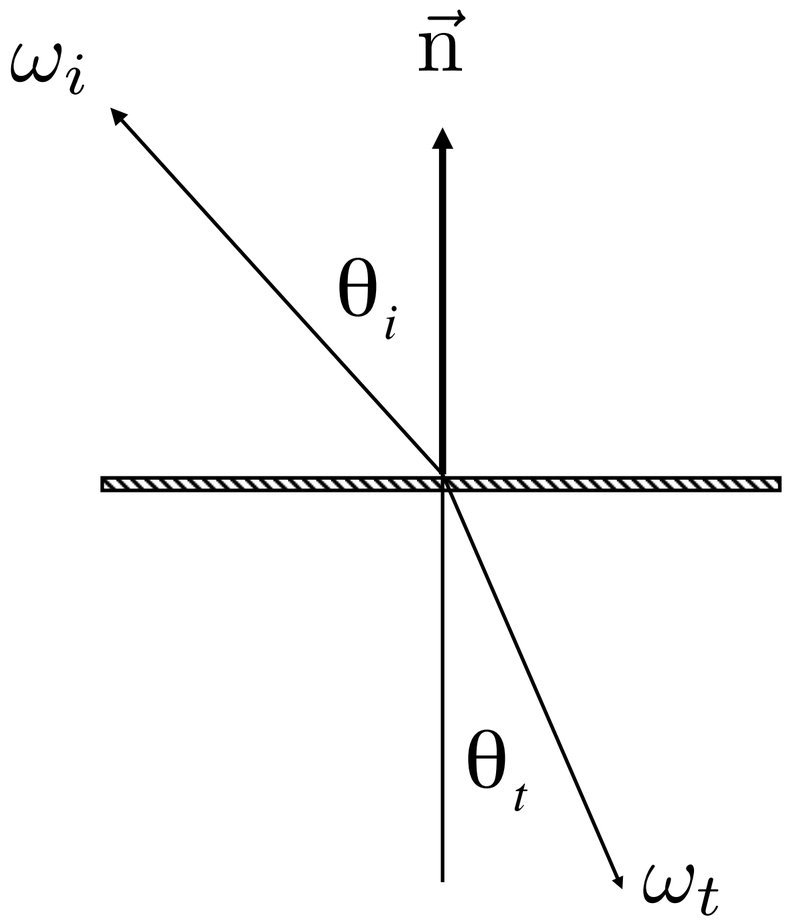
Snell's Law

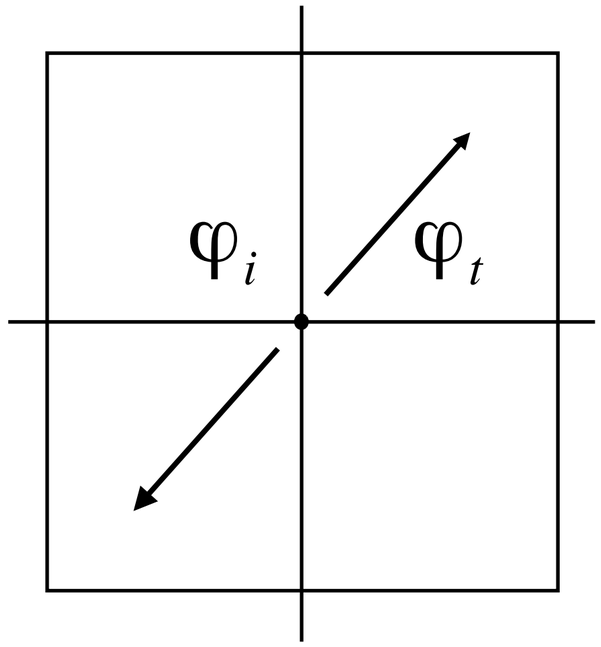
Transmitted angle depends on

index of refraction (IOR) for incident ray

index of refraction (IOR) for exiting ray



$$\eta_i \sin\theta_i = \eta_t \sin\theta_t$$

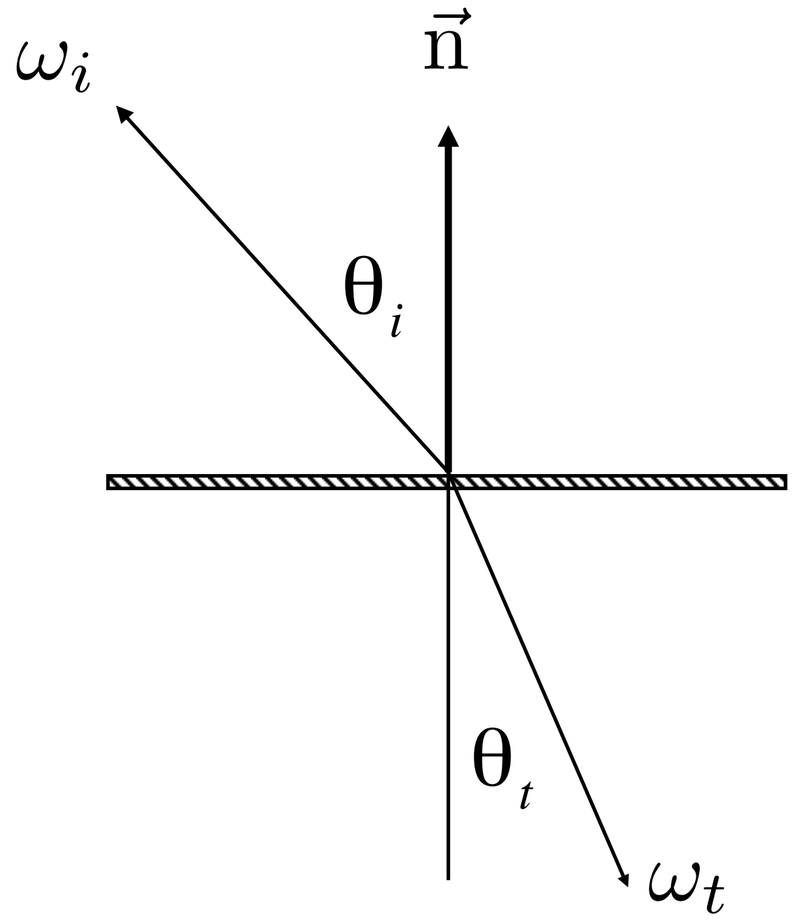


$$\varphi_t = \varphi_i \pm \pi$$

Medium	η*
Vacuum	1.0
Air (sea level)	1.00029
Water (20°C)	1.333
Glass	1.5-1.6
Diamond	2.42

* index of refraction is wavelength dependent (these are averages)

Law of Refraction



$$\eta_i \sin\theta_i = \eta_t \sin\theta_t$$

$$\cos\theta_t = \sqrt{1 - \sin^2\theta_t}$$

$$= \sqrt{1 - \left(\frac{\eta_i}{\eta_t}\right)^2 \sin^2\theta_i}$$

$$= \sqrt{1 - \left(\frac{\eta_i}{\eta_t}\right)^2 (1 - \cos^2\ \theta_i)}$$

Total internal reflection:

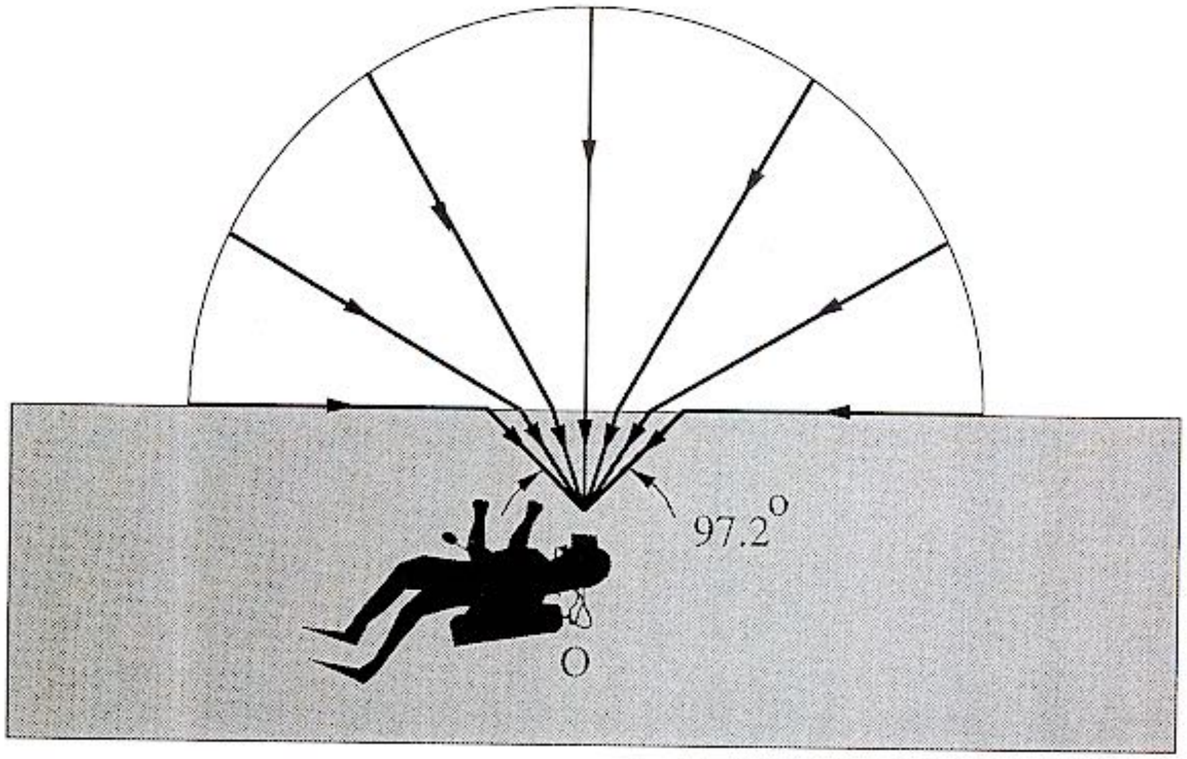
$$1 - \left(\frac{\eta_i}{\eta_t}\right)^2 (1 - \cos^2\theta_i) < 0$$

When light is moving from a more optically dense medium to a less optically dense medium: $\frac{\eta_i}{\eta_t} > 1$

Light incident on boundary from large enough angle will not exit medium.

Snell's Window/Circle

Total internal reflection



CS184 [Livingston and Lynch] Ren Ng

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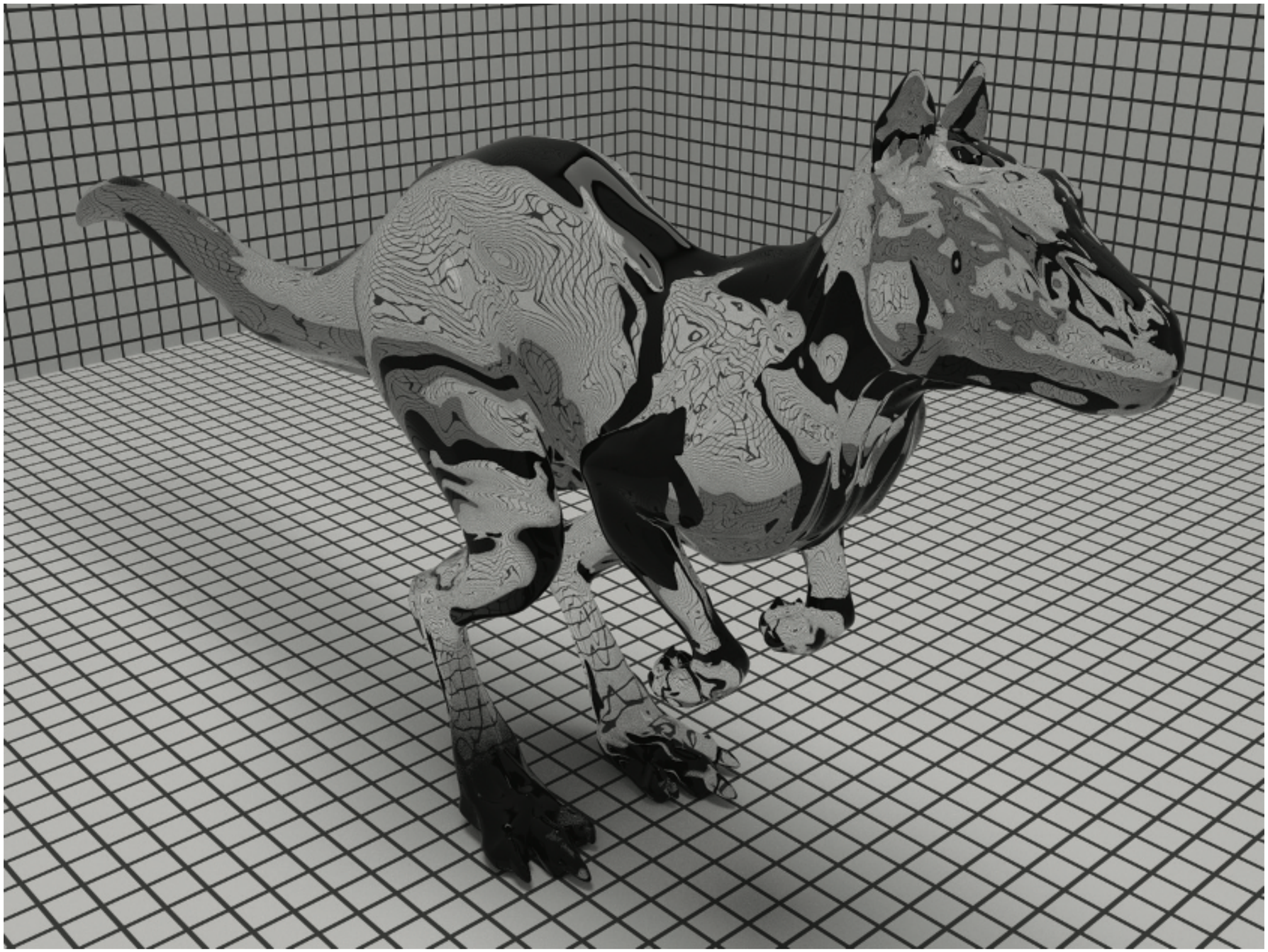
Fresnel Reflection & Transmission



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Microfacet Material Model

Microfacet Reflection

https://twitter.com/Cmdr_Hadfield/status/318986491063828480/photo/1

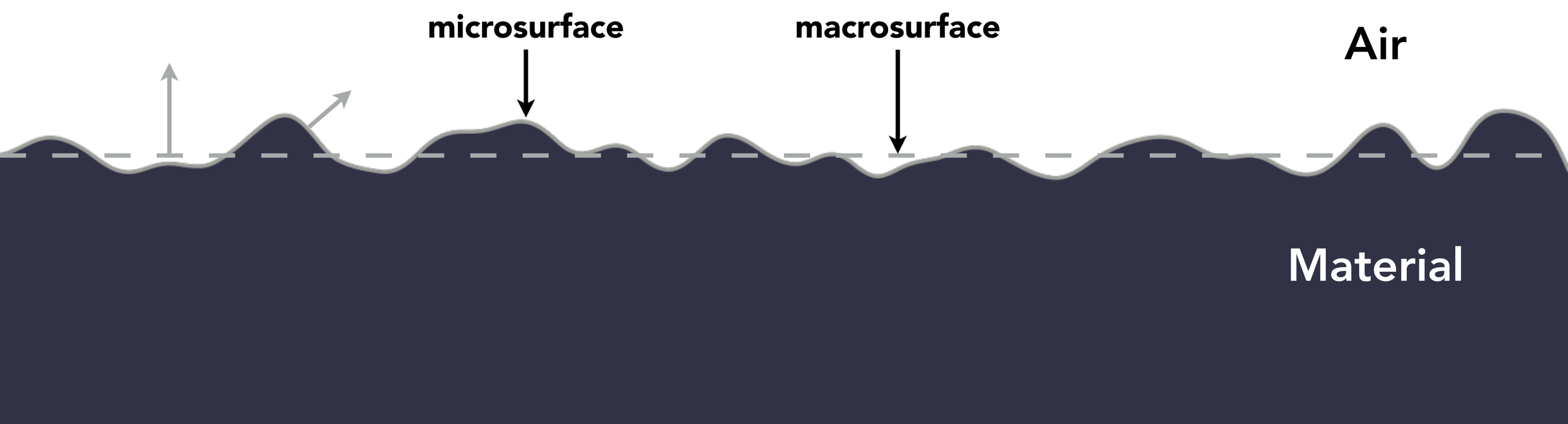
Microfacet Theory

Rough surface

- Macroscale: flat & rough
- Microscale: bumpy & specular

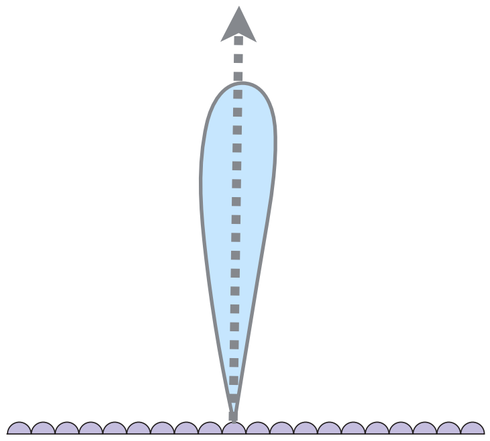
Individual elements of surface act like mirrors

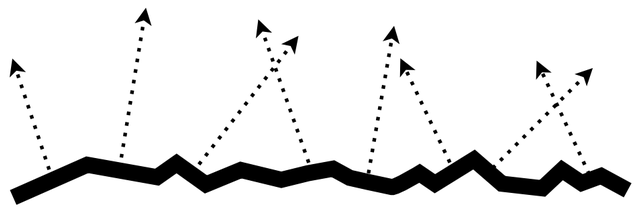
- Known as “microfacets”
- Each microfacet has its own normal vector

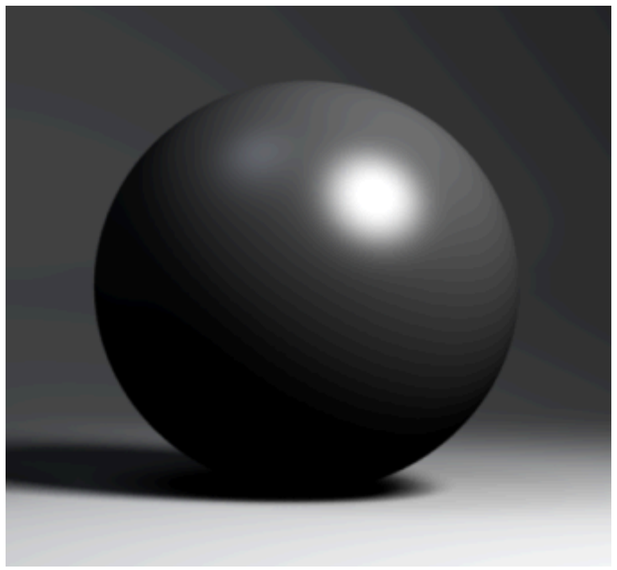


Microfacet BRDF

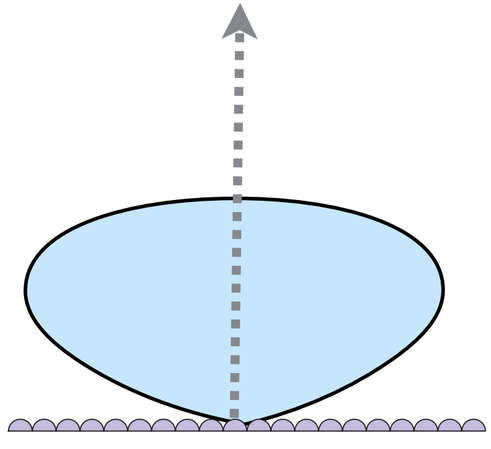
- Key: the distribution of microfacets' normals
  - Concentrated <==> glossy

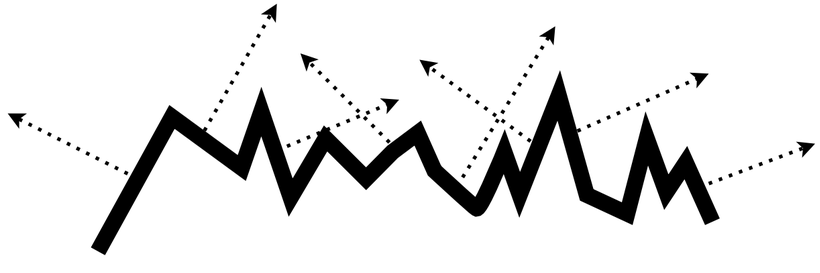


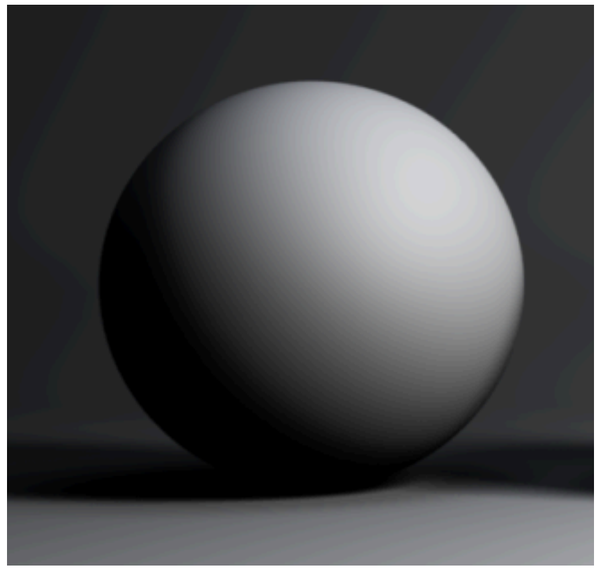




  - Spread out <==> diffuse

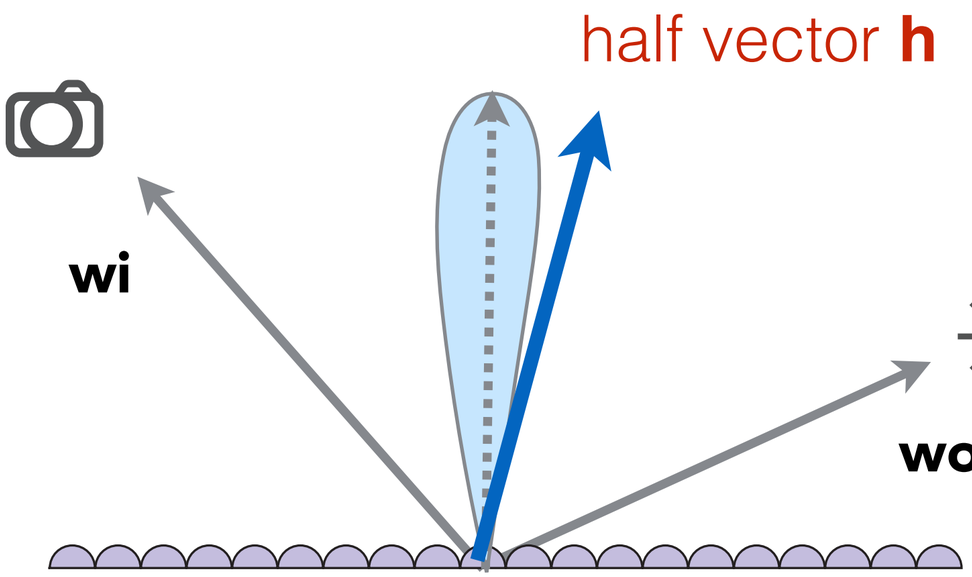






Microfacet BRDF

- What kind of microfacets reflect wi to wo? (hint: microfacets are mirrors)



Fresnel term | shadowing-masking term | **distribution of normals**

$$f(\mathbf{i},\mathbf{o}) = \frac{\mathbf{F}(\mathbf{i},\mathbf{h})\mathbf{G}(\mathbf{i},\mathbf{o},\mathbf{h})\mathbf{D}(\mathbf{h})}{4(\mathbf{n},\mathbf{i})(\mathbf{n},\mathbf{o})}$$

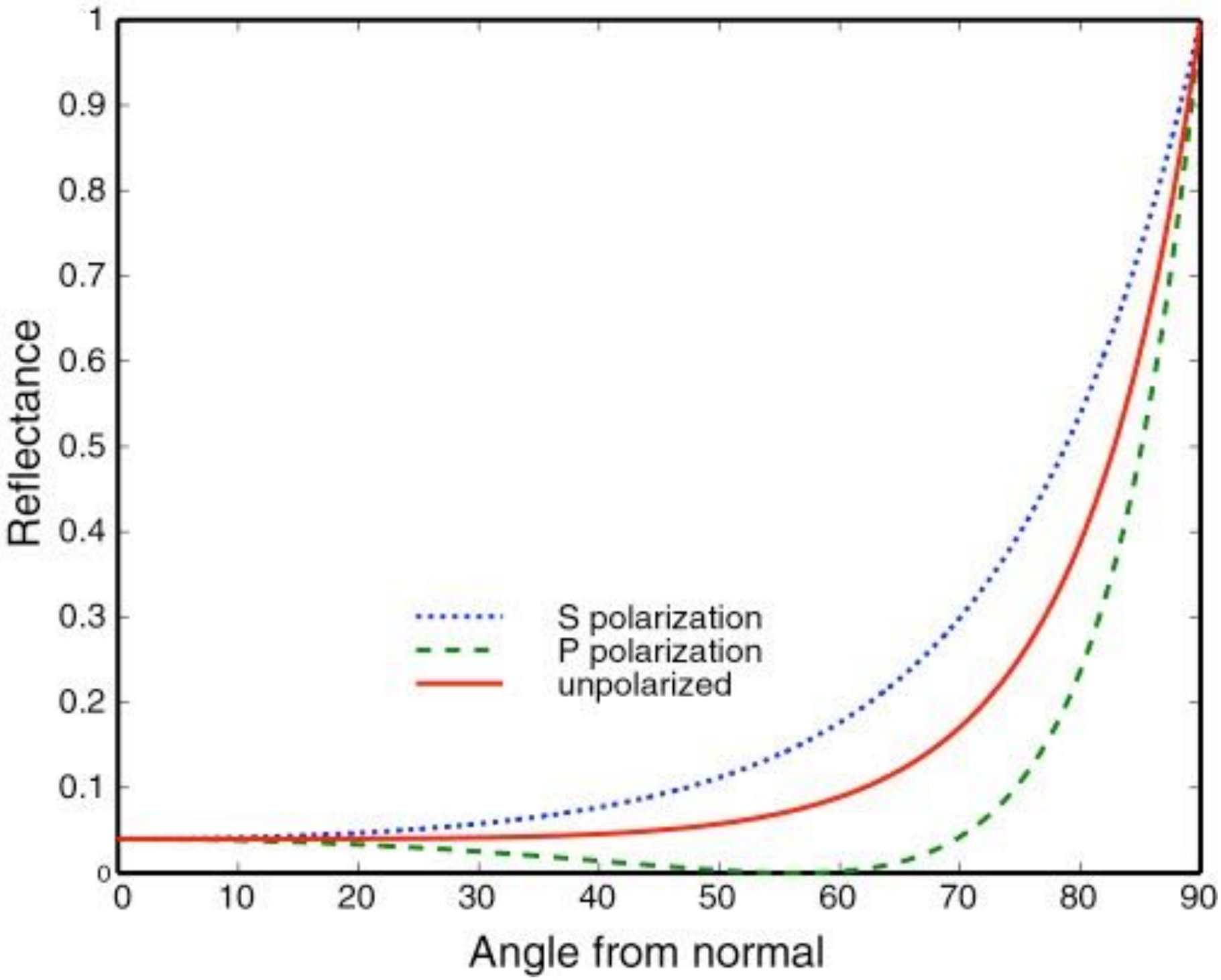
Fresnel Reflection Term

Reflectance depends on incident angle (and polarization of light)

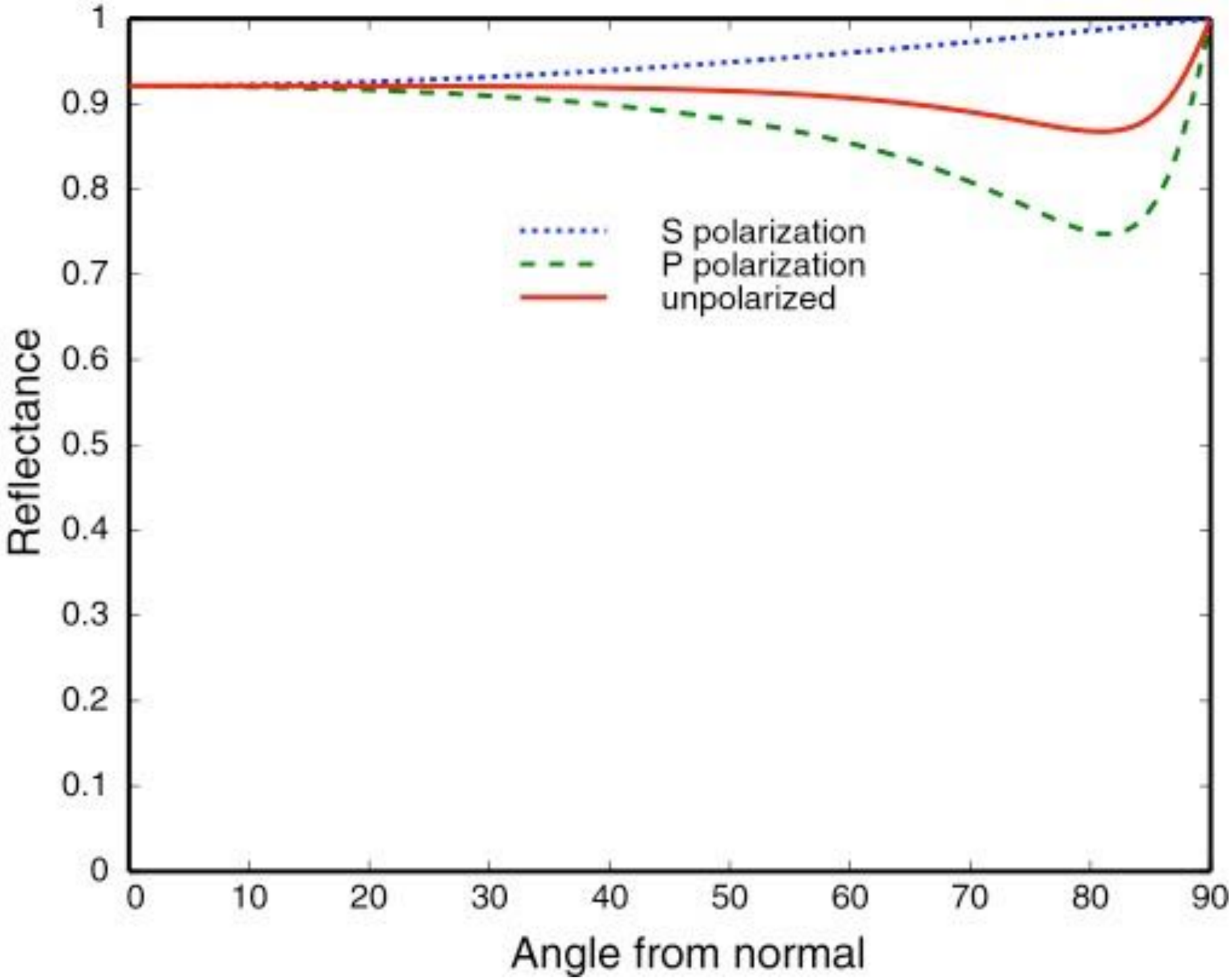
This example: reflectance increases with grazing angle

[Lafortune et al. 1997]

Fresnel Term (Dielectric, η =1.5)



Fresnel Term (Conductor)



Microfacet BRDF: Examples

[Autodesk Fusion 360]

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Attendance Time

If you are seated in class, go to this form and sign in:

- https://bit.ly/184attendance

Notes:

- Time-stamp will be taken when you submit form.
  Do it now, won't count later.
- Don't tell friends outside class to fill it out now, because we will audit at some point in semester.
- Failing audit will have large negative consequence. You don't need to, because you have an alternative!

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Anisotropic BRDFs

Isotropic vs Anisotropic Reflection

- So far, Point light + Metal = Round / Elliptical highlight
- But some reflection highlights look very different

Isotropic vs Anisotropic Reflection

Isotropic

Anisotropic

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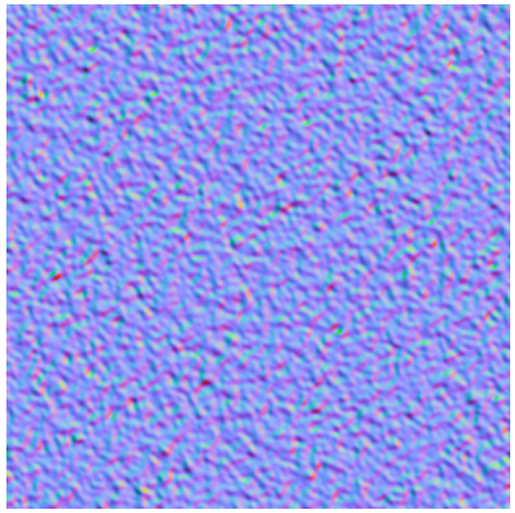
Anisotropic BRDF: Brushed Metal

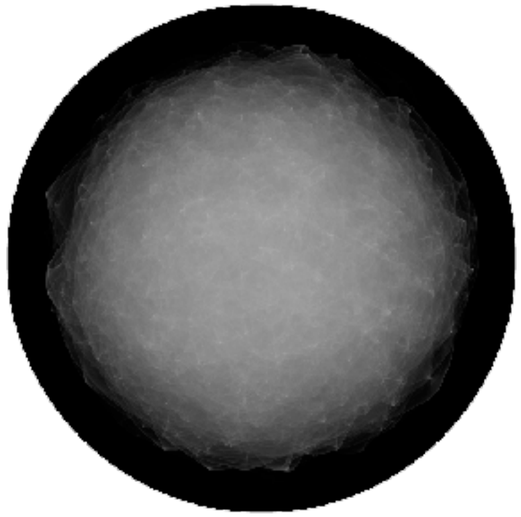
- How is the pan brushed?

Isotropic / Anisotropic Materials (BRDFs)

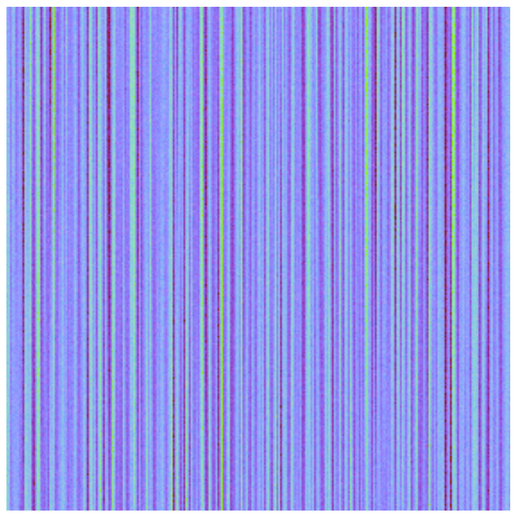
- Key: directionality of underlying surface

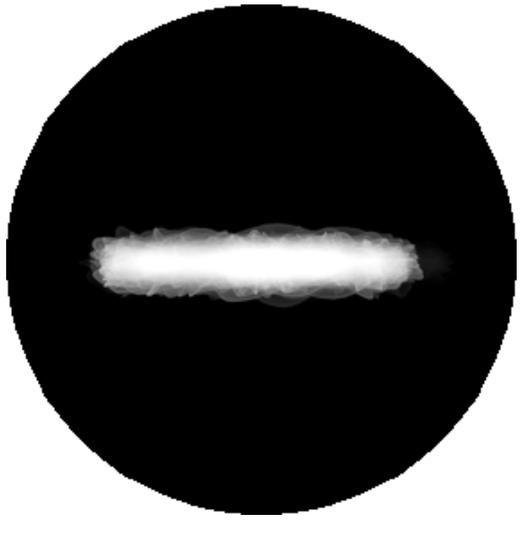
Isotropic





Anisotropic





Surface (normals)

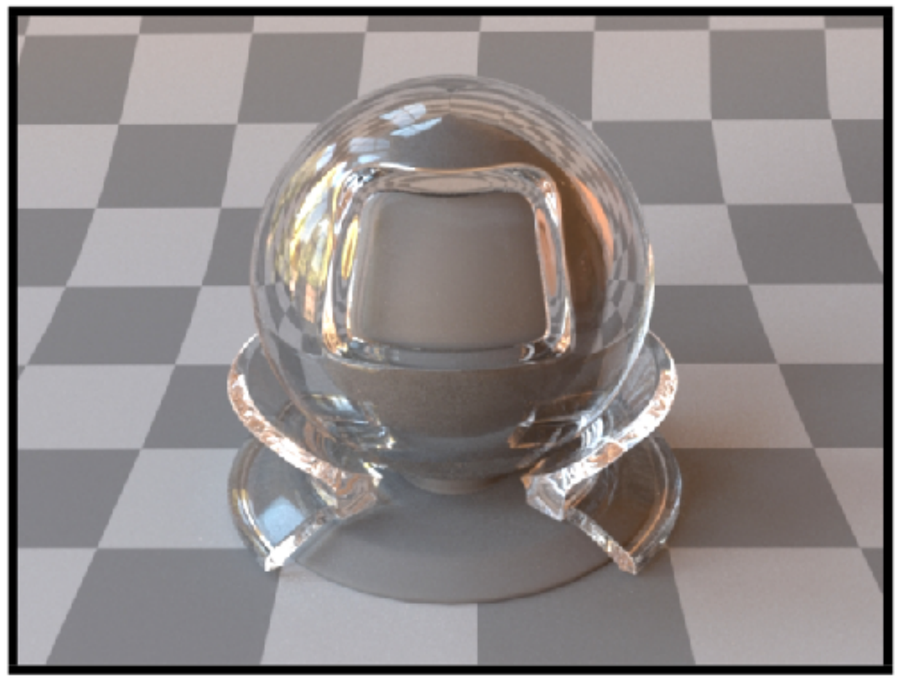
BRDF (fix wi, vary wo)

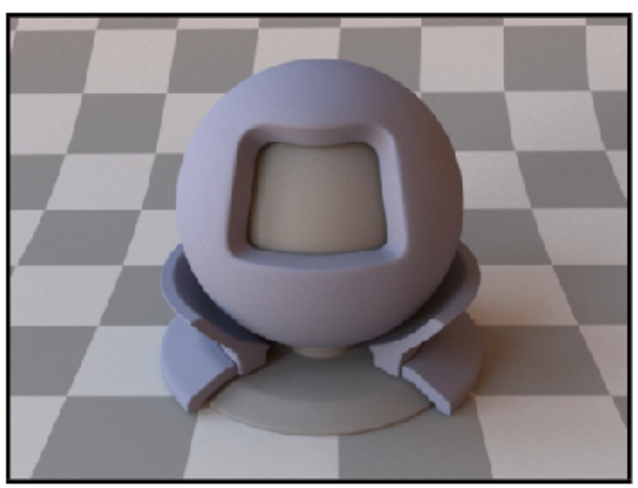
Isotropic BRDFs

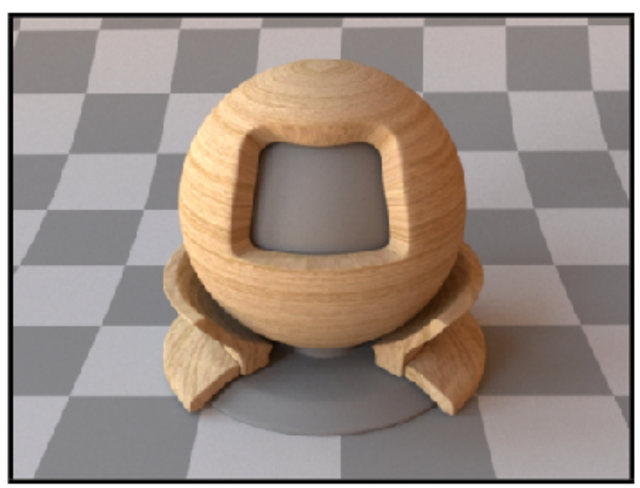
Reflection independent of azimuthal angle ϕ

$$f_r(\theta_i, \phi_i; \theta_r, \phi_r) = f_r(\theta_i, \theta_r, \phi_r - \phi_i)$$

Results from surface microstructure that lacks directional structure







Ren Ng

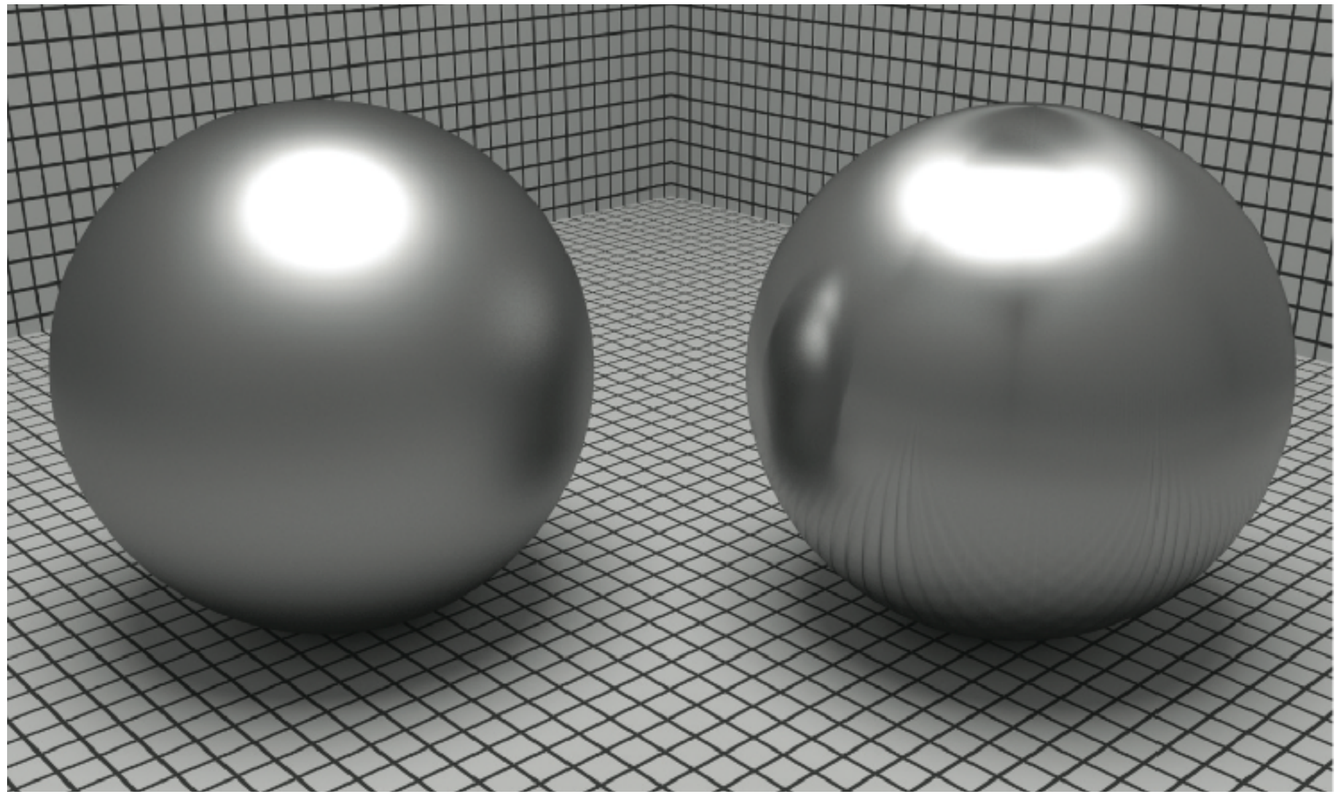
Presented by David McAllister

Anisotropic BRDFs

Reflection depends on azimuthal angle ϕ

$$f_r(\theta_i, \phi_i; \theta_r, \phi_r) \neq f_r(\theta_i, \theta_r, \phi_r - \phi_i)$$

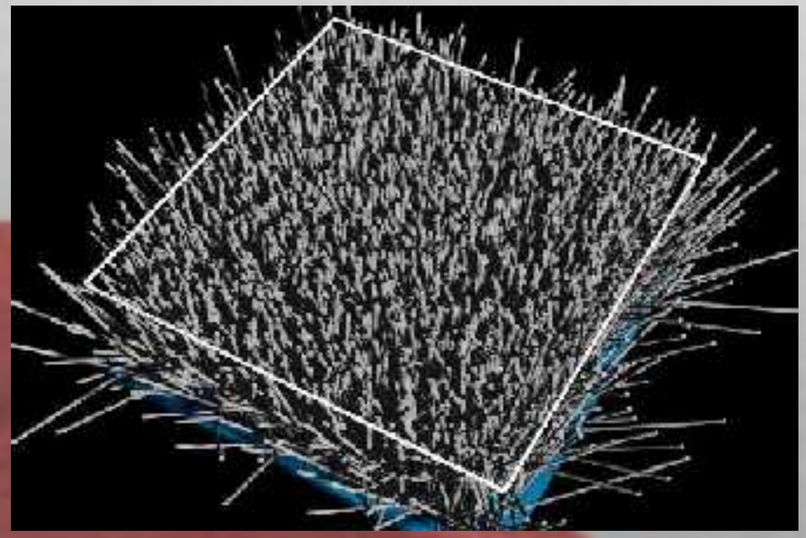
Results from oriented microstructure of surface, e.g., brushed metal



Anisotropic BRDF: Nylon

[Westin et al. 1992]

Anisotropic BRDF: Velvet



[Westin et al. 1992]

Anisotropic BRDF: Velvet

[https://www.youtube.com/watch?v=2hjoW8TYTd4]

Sampling of Advanced Material Modeling Topics (Slides courtesy Prof Lingqi Yan)

Detailed / Glinty Material

Why details?

Microfacet model

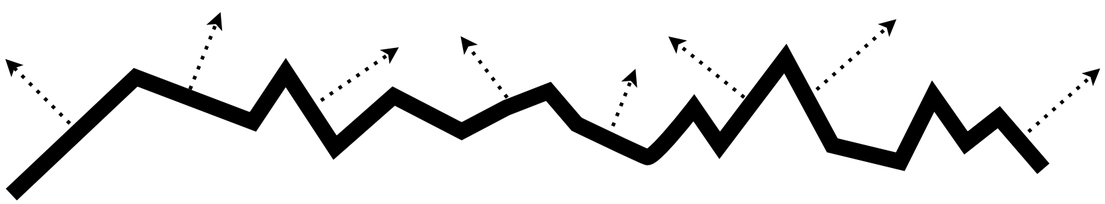
Why details?

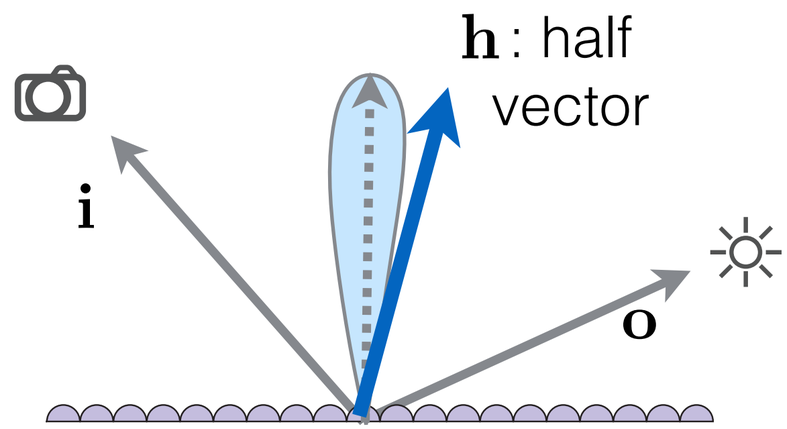
[Yan et al. 2014, 2016]

Why details?

[Yan et al. 2014, 2016]

Recap: Microfacet BRDF



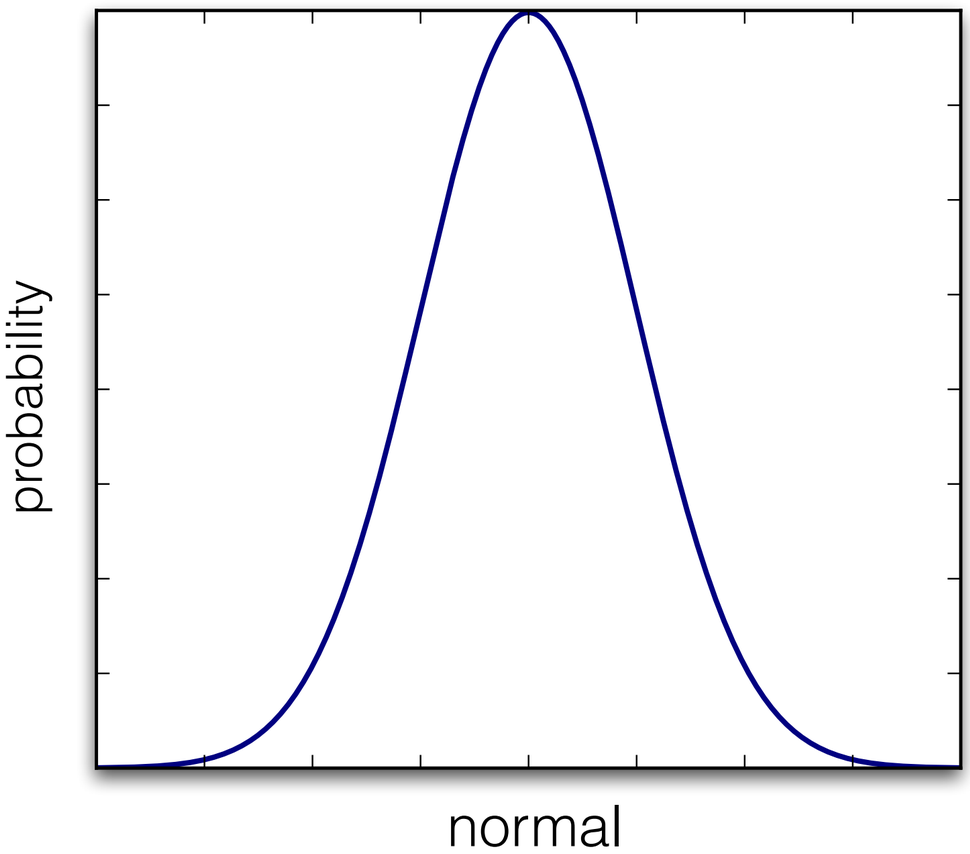


Surface = Specular microfacets + statistical normals

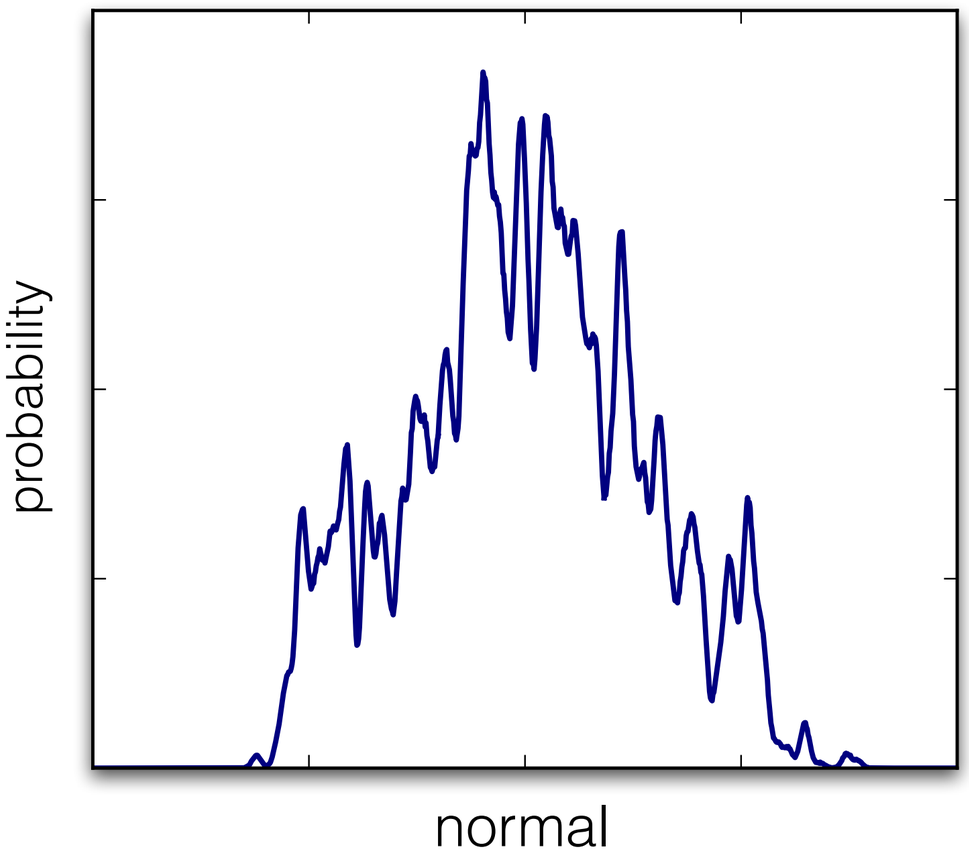
$$f(\mathbf{i},\mathbf{o}) = \frac{\mathbf{F}(\mathbf{i},\mathbf{h})\mathbf{G}(\mathbf{i},\mathbf{o},\mathbf{h})\mathbf{D}(\mathbf{h})}{4(\mathbf{n},\mathbf{i})(\mathbf{n},\mathbf{o})}$$

NDF: Normal Distribution Function

Statistical NDF vs. Actual NDF

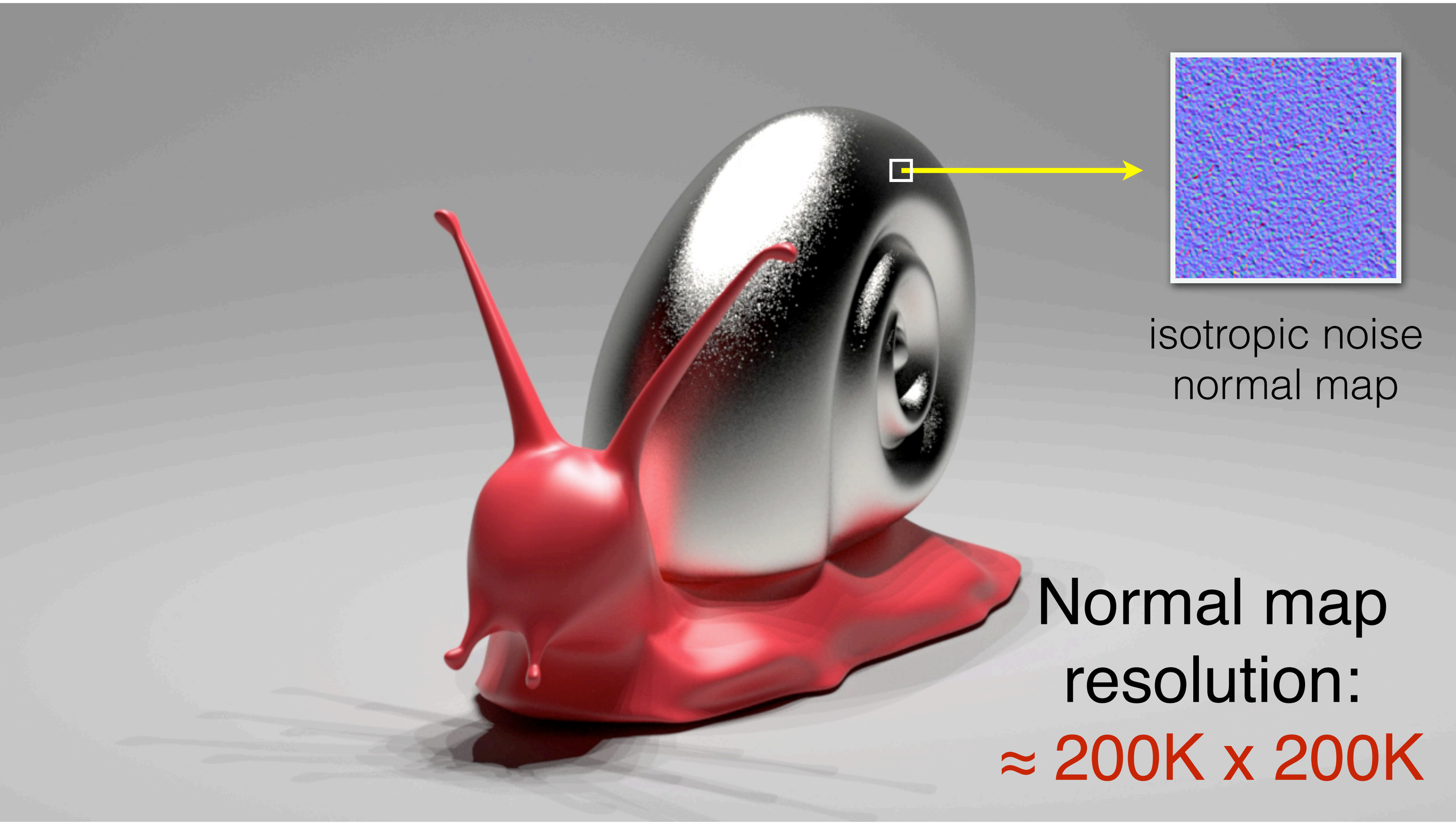


What we have
(microfacet — statistical)



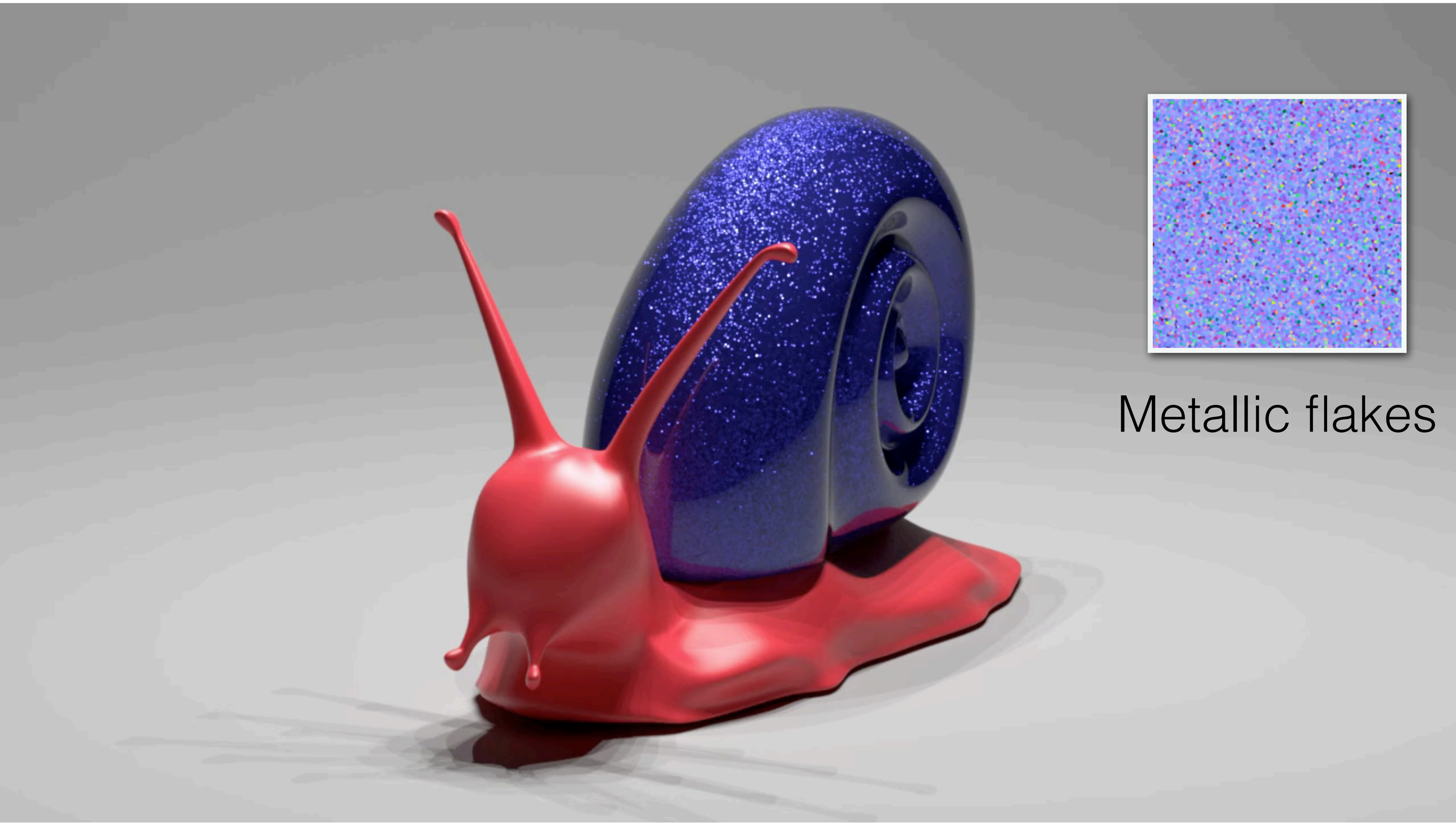
What we want

Define Details



Ren Ng
Presented by David McAllister

Different Details



Metallic flakes

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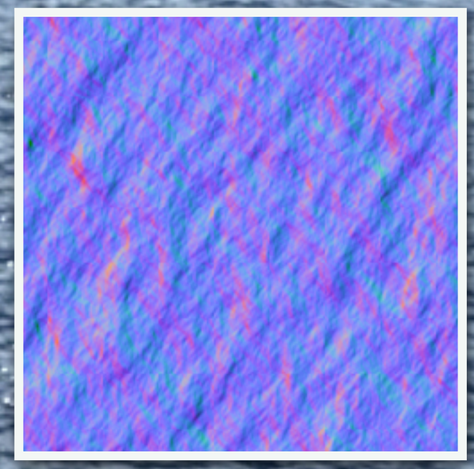
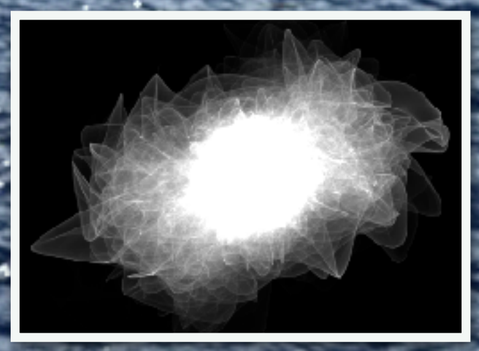
Presented by David McAllister





Ocean waves

Hair / Fur Appearance Models

Hair Appearance

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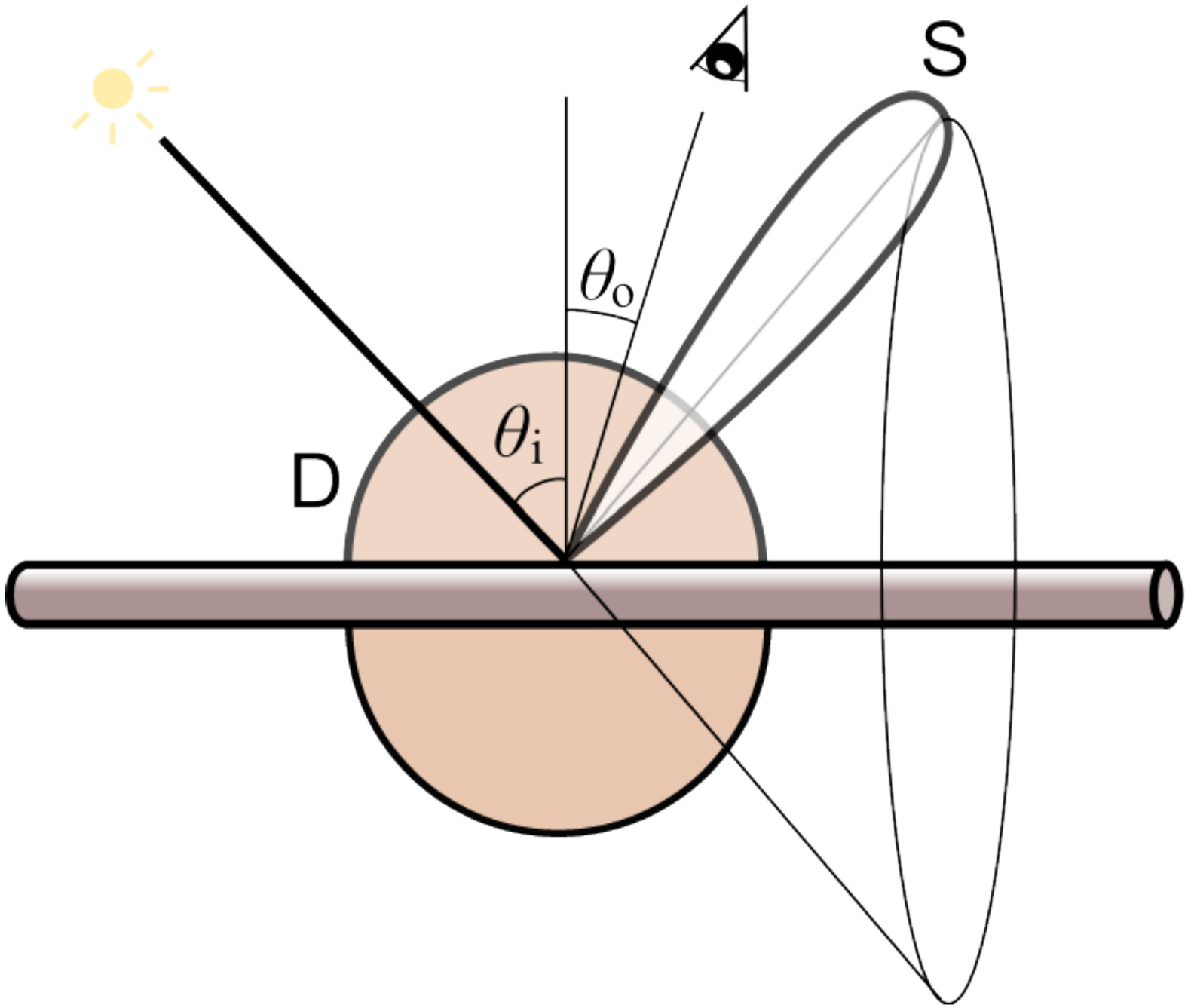
Kajiya-Kay Model



[Image courtesy of Chiwei Tseng]

Kajiya-Kay Model

[Yuksel et al. 2008]

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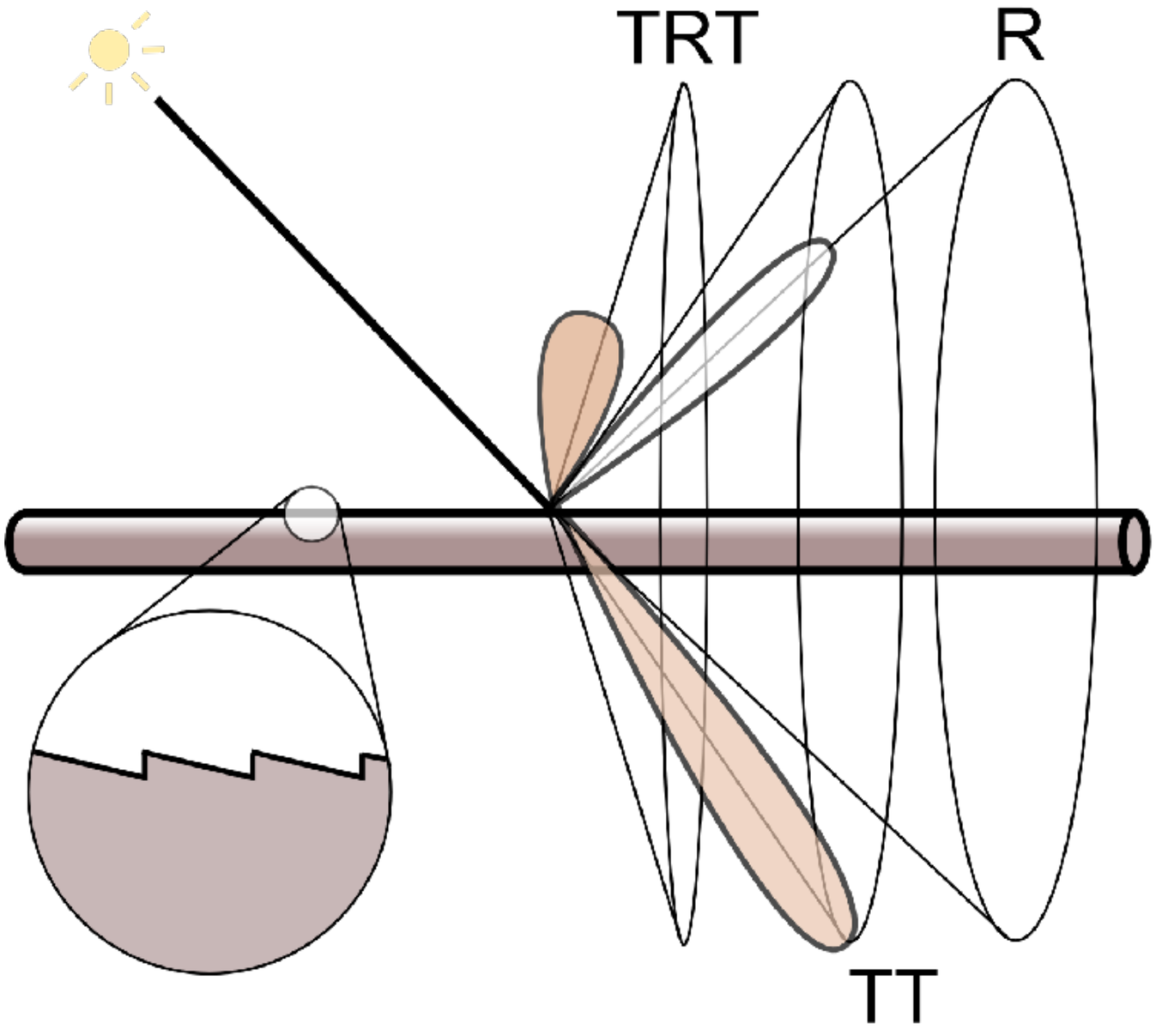
Presented by David McAllister

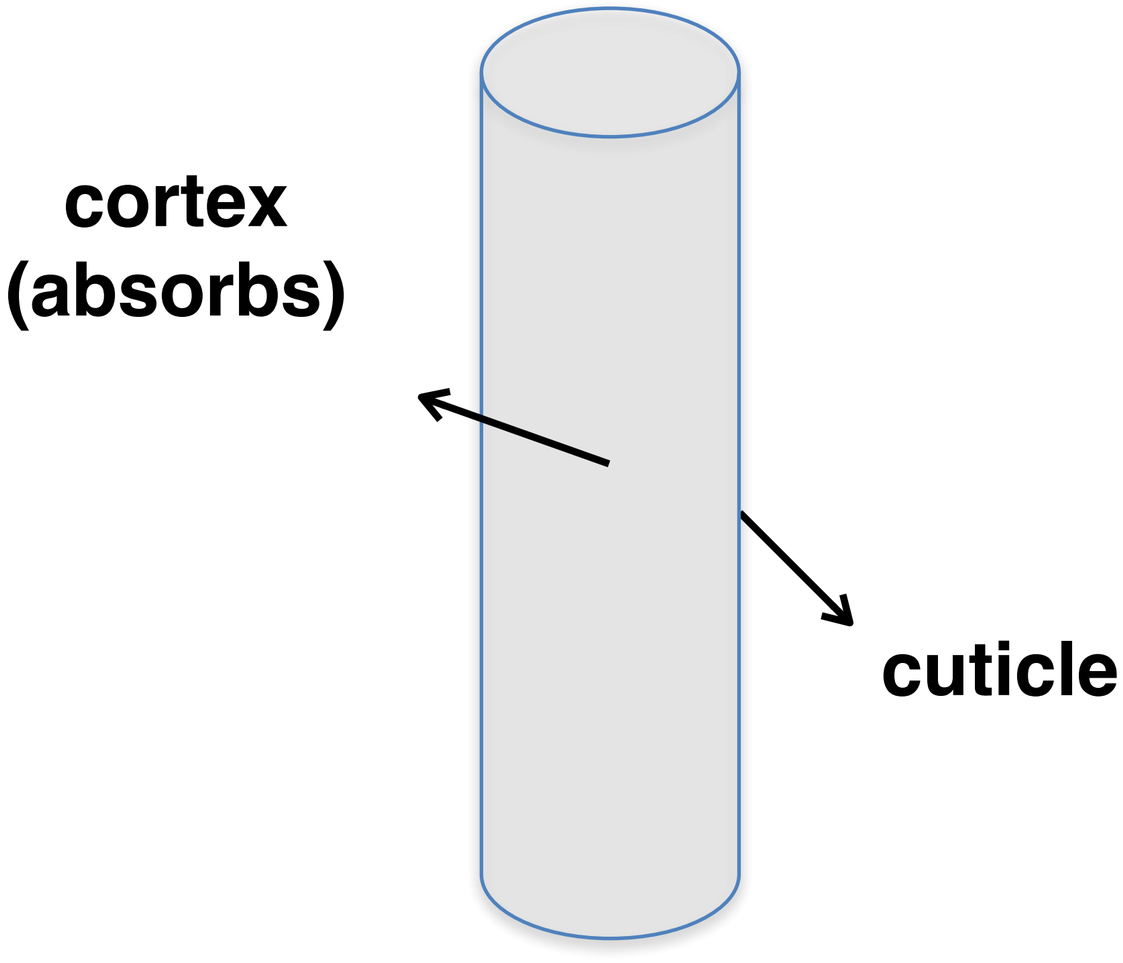
Marschner Model



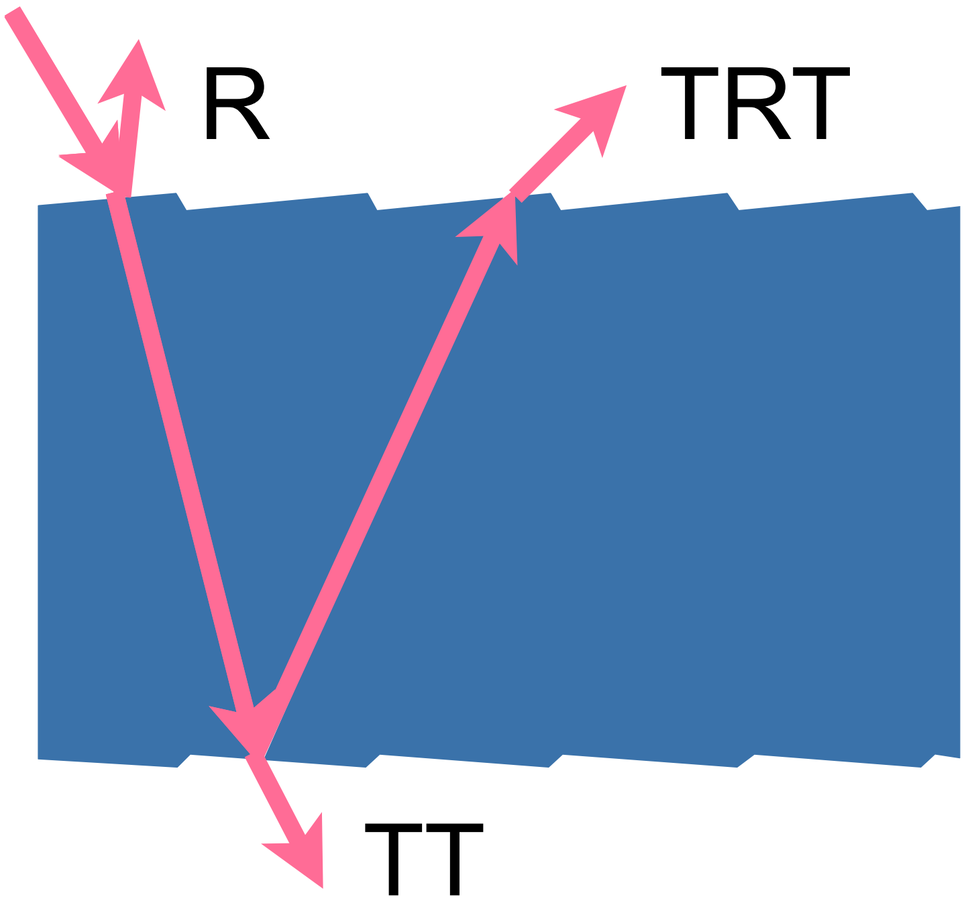
[Image courtesy of Chiwei Tseng]

Marschner Model

- Glass-like cylinder



- 3 types of light interactions:
  R, TT, TRT
  (R: reflection, T: transmission)



[Marschner et al. 2003]

Marschner model

[Marschner et al. 2003]

[d'Eon et al. 2011]

Participating Media

Participating Media: Fog

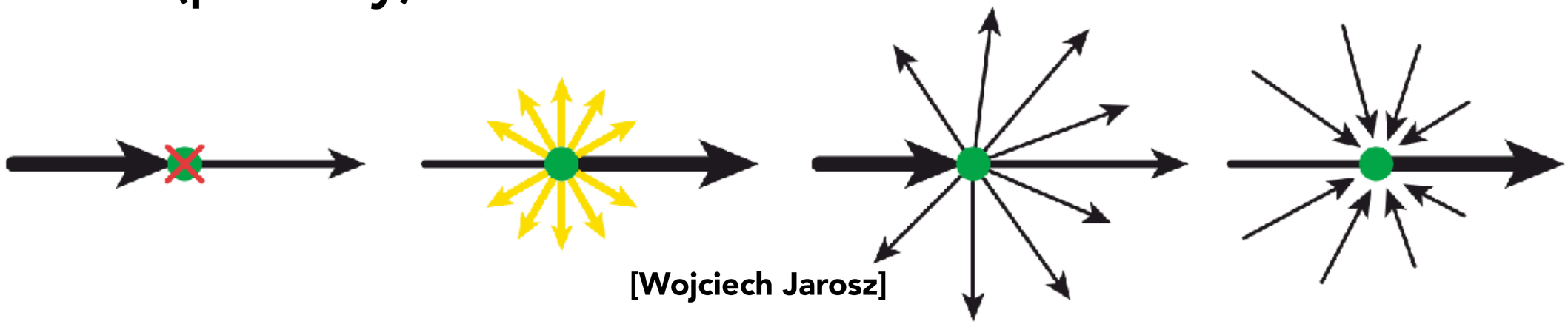
[Novák et al. 2012]

Participating Media: Cloud

[by thephotographer0]

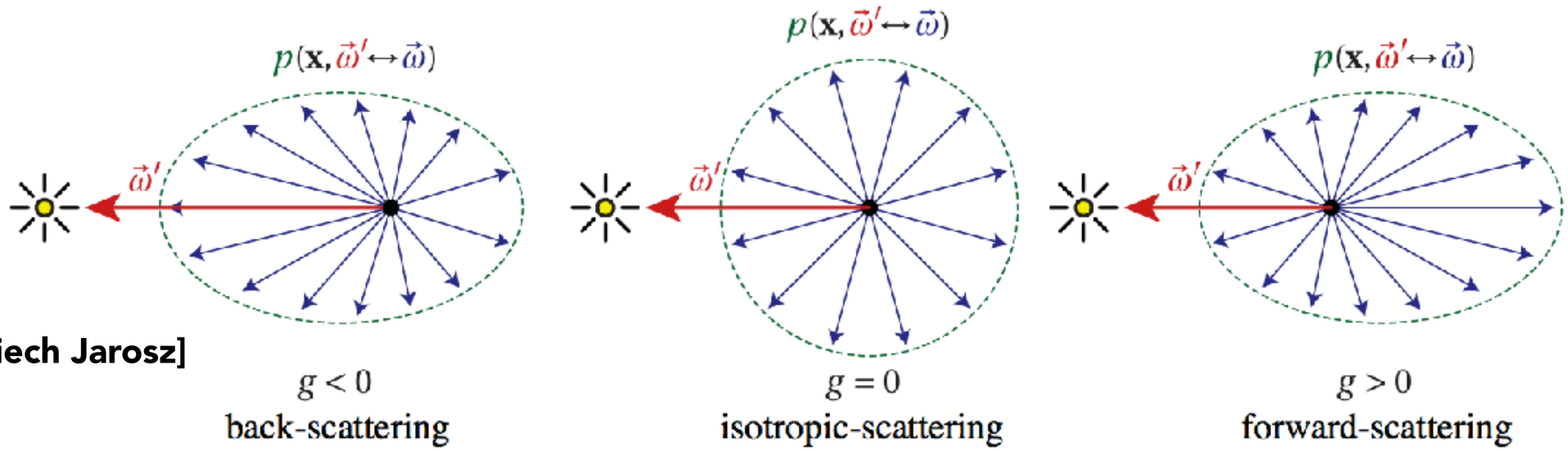
Participating Media

- At any point as light travels through a participating medium, it can be (partially) absorbed and scattered.



[Wojciech Jarosz]

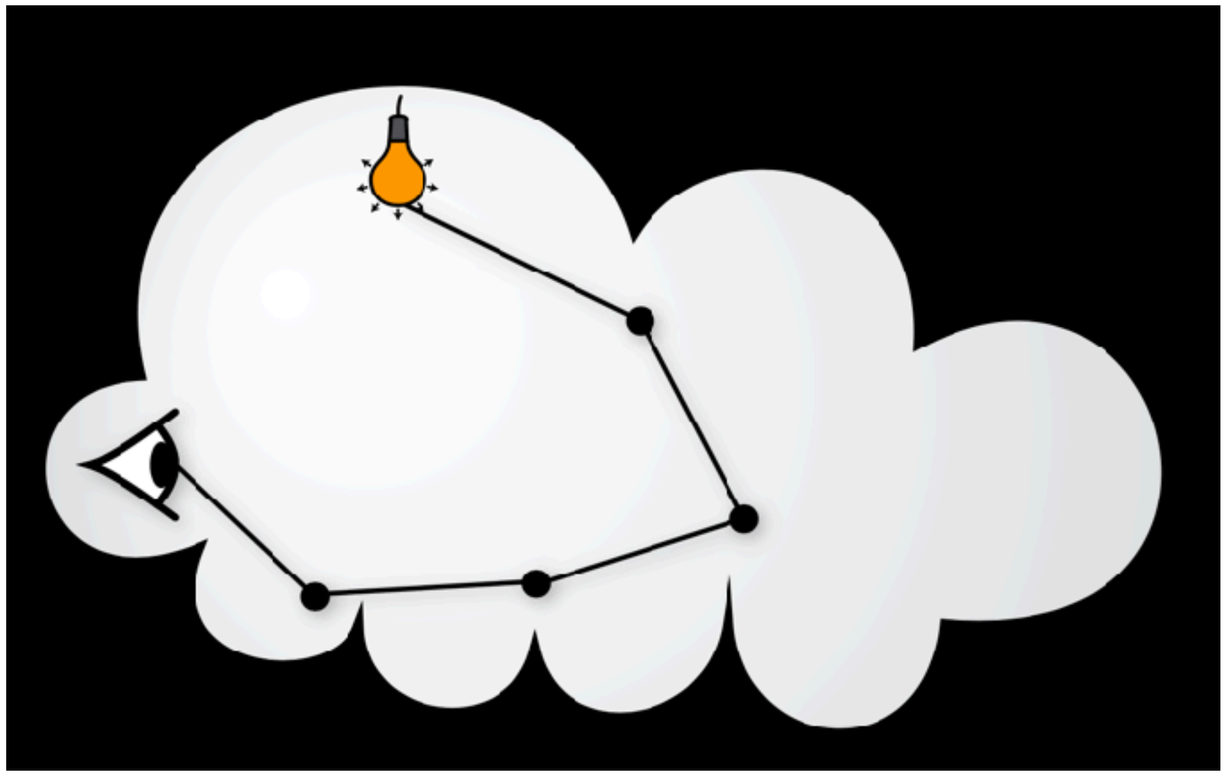
- Use Phase Function to describe the angular distribution of light scattering at any point x within participating media.

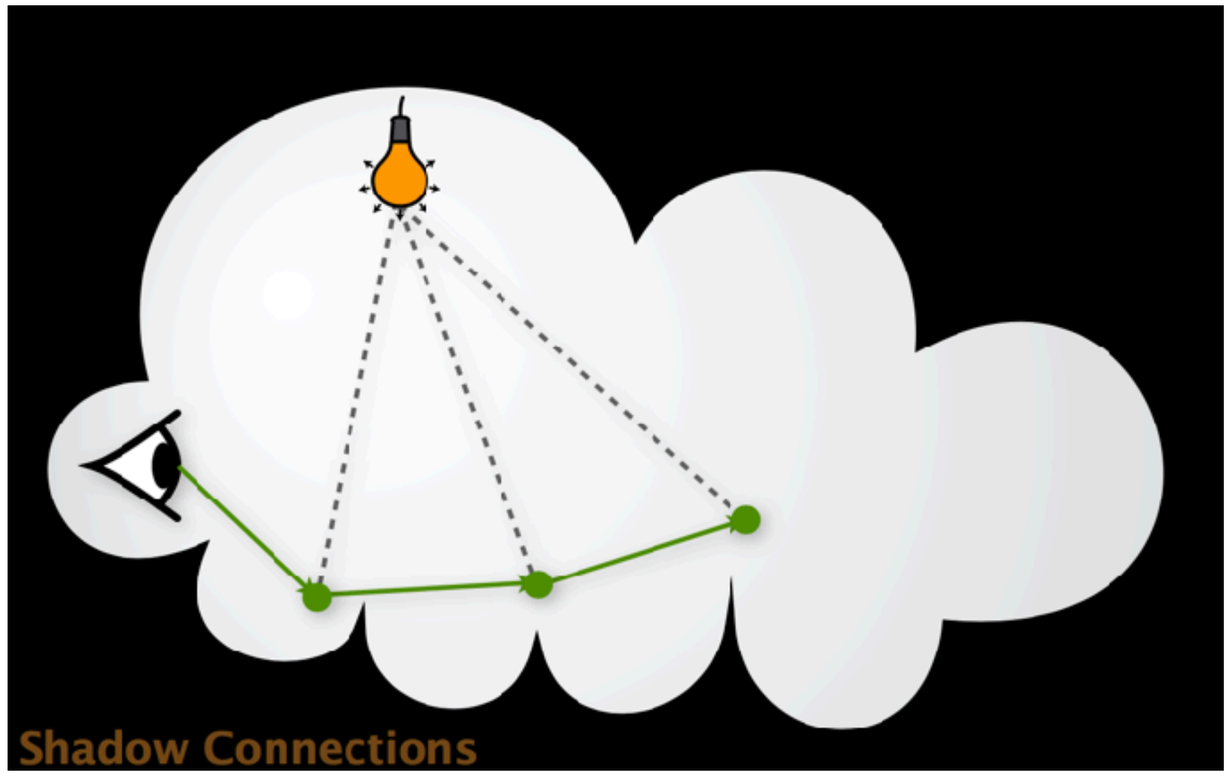


[Wojciech Jarosz]

Participating Media: Rendering

- Randomly choose a direction to bounce
- Randomly choose a distance to go straight
- At each ‘shading point’, connect to the light





[Derek Nowrouzezahrai]

Participating Media: Application

[Big Hero 6, 2014 Disney]

Participating Media: Application

[Assassin's Creed Syndicate. 2015 Ubisoft]

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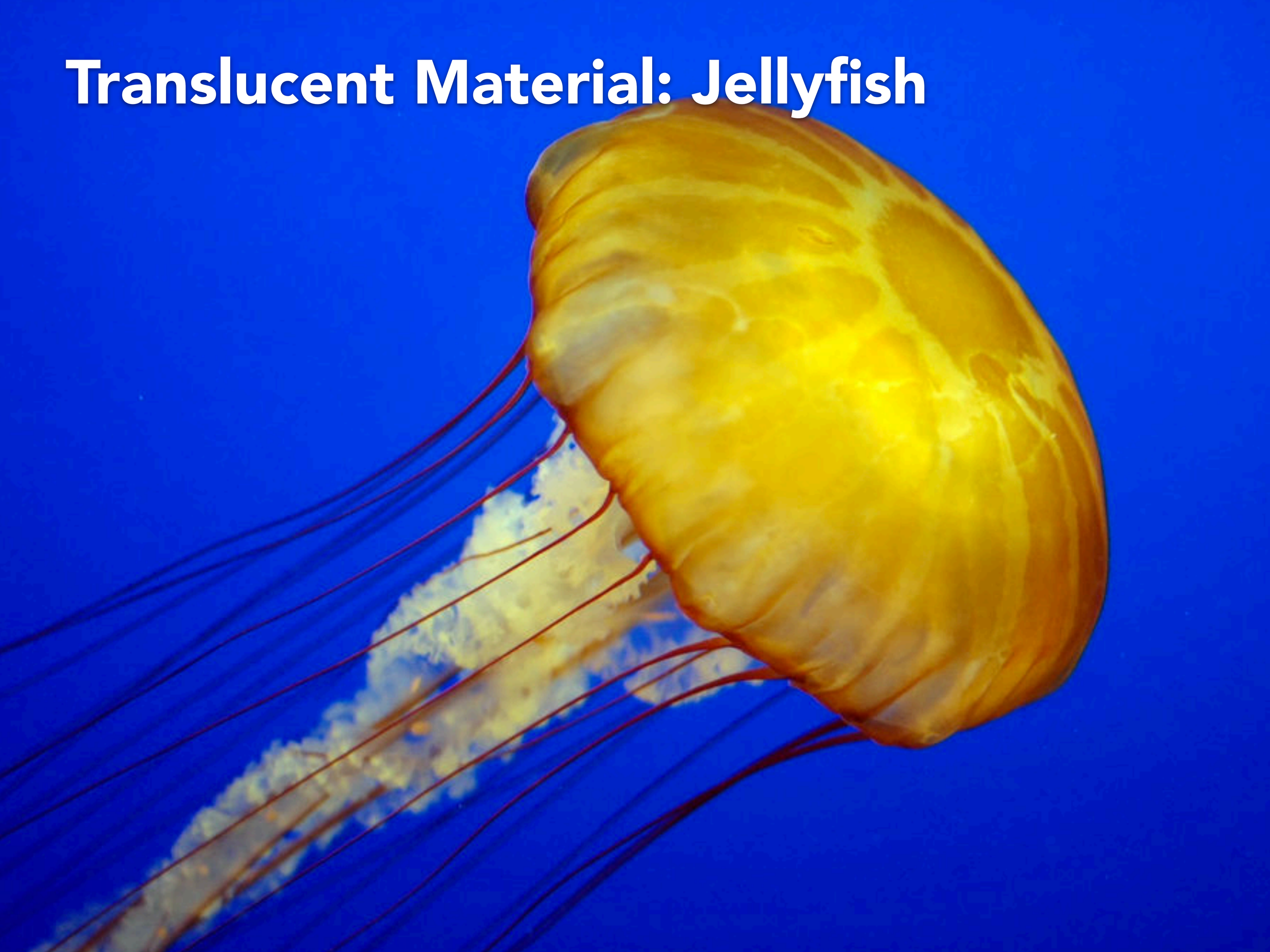
Translucent Material (specific participating media)

Translucent Material: Jade

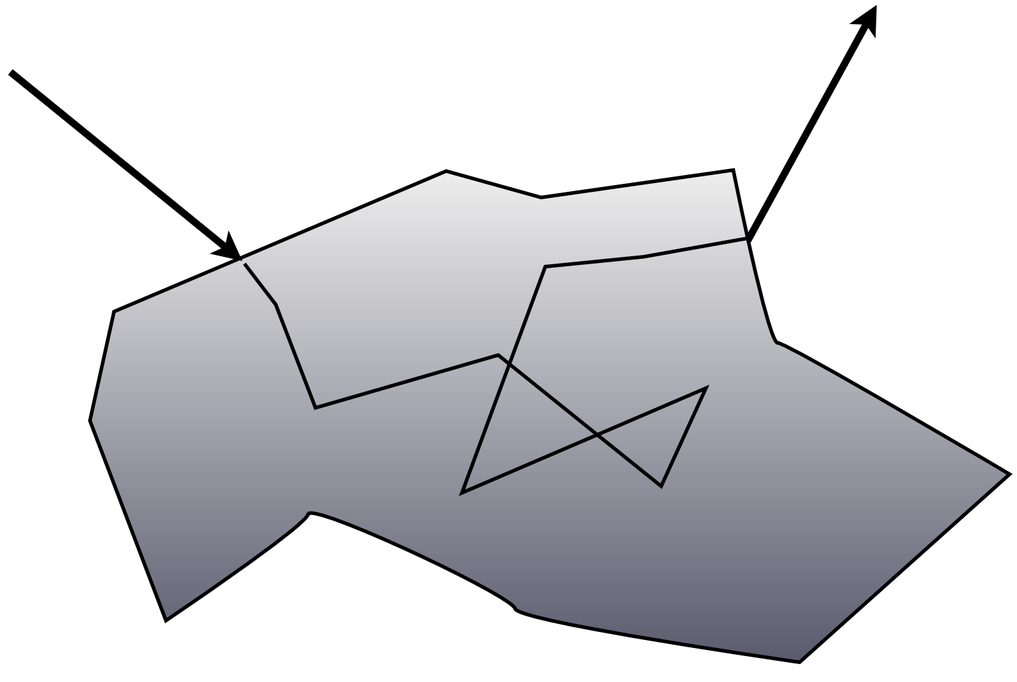
Translucent Material: Jellyfish



Subsurface Scattering

Visual characteristics of many surfaces caused by light exiting at different points than it enters

- Violates a fundamental assumption of the BRDF



- Different from transparent

[Jensen et al 2001]

[Donner et al 2008]

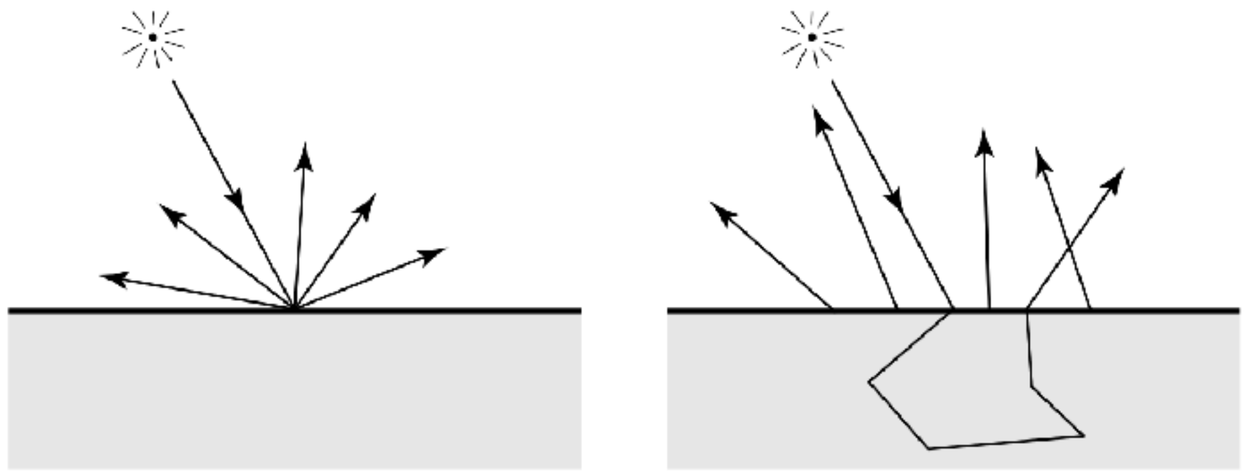
Scattering Functions

- BSSRDF: generalization of BRDF; exitant radiance at one point due to incident differential irradiance at another point:

$$S(x_i, \omega_i, x_o, \omega_o)$$

- Generalization of rendering equation: integrating over all points on the surface and all directions (!)

$$L(x_o, \omega_o) = \int_A \int_{H^2} S(x_i, \omega_i, x_o, \omega_o)\, L_i(x_i, \omega_i) \cos\theta_i \, \mathrm{d}\omega_i \, \mathrm{d}A$$



BRDF

BSSRDF

[Jensen et al. 2001]

BRDF

[Jensen et al. 2001]

BSSRDF

[Jensen et al. 2001]

BRDF vs BSSRDF

BRDF

BSSRDF

[Jensen et al. 2001]

Attendance Time

If you are seated in class, go to this form and sign in:

- https://forms.gle/zFdQh9gBwa1sVMLa6
- Attendance word: bidirectional

Notes:

- Time-stamp will be taken when you submit form.
  Do it now, won't count later.
- Don't tell friends outside class to fill it out now, because we will audit at some point in semester.
- Failing audit will have large negative consequence. You don't need to, because you have an alternative!

Acknowledgments

Thanks to Lingqi Yan, Matt Pharr, Pat Hanrahan and Kayvon Fatahalian for slide resources.

Extra: Inverse Rendering

Recovering

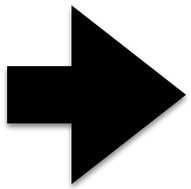
geometry**, **materials**, **lighting, and cameras

from images

What is inverse rendering?

Forward rendering (this class)

Geometry
Materials
Lighting
Cameras

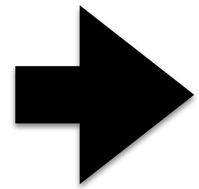


Rendered images

Use physically accurate models to render pretty images!

Inverse rendering

Captures images



Geometry
Materials
Lighting
Cameras

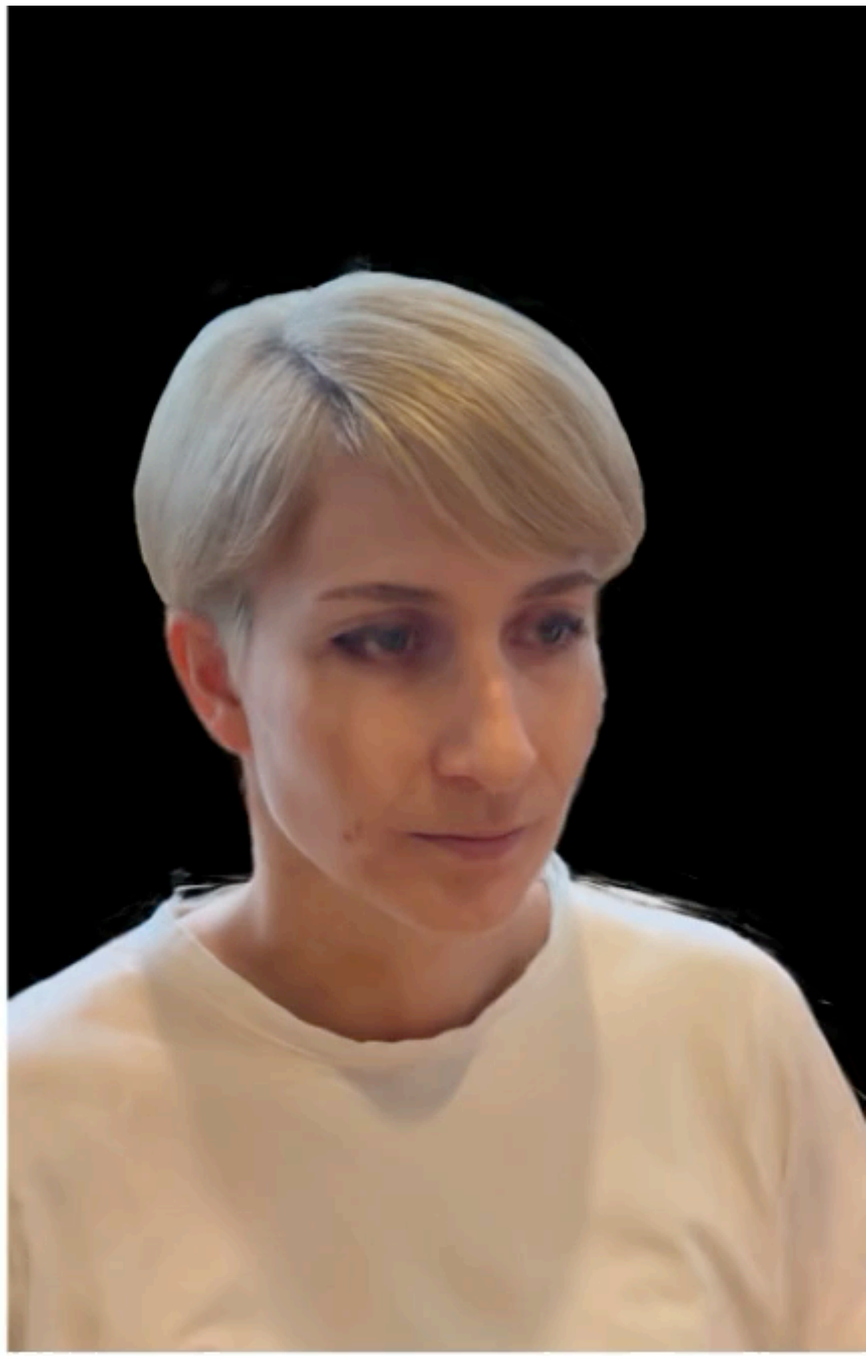
Recover the underlying properties from multiple observed images!

NeRFs and Gaussian Splatting are common techniques!

Inverse Rendering for Hair

Input video

Reconstruction



Rendering

CS184

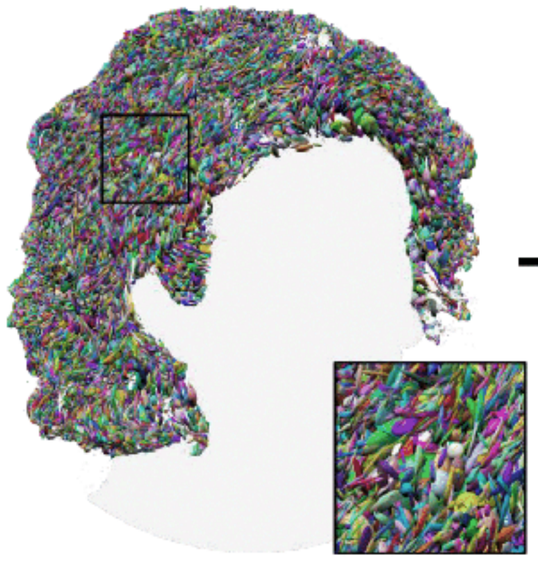
"Gaussian Haircut ✂️", ECCV 2024

Presented by Da

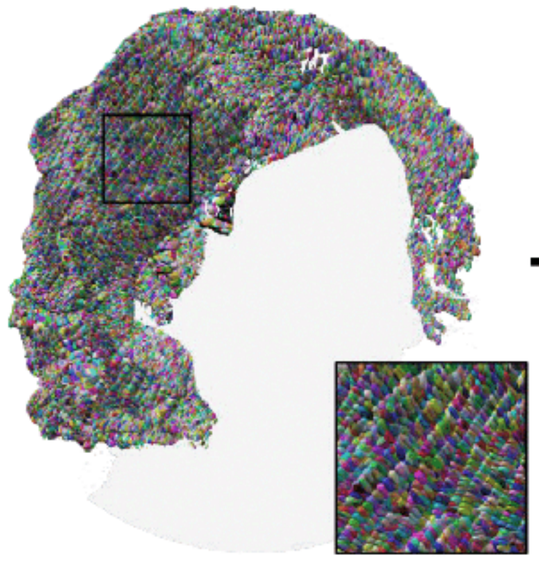
Inverse Rendering for Hair

Multi-view images

Stage 1: 3D Lifting
Unstructured Gaussians



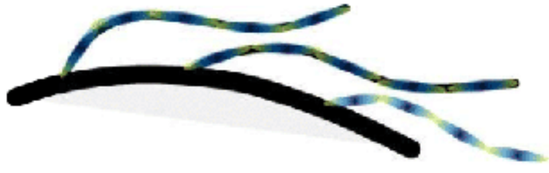
Stage 2: Strands fitting
Structured Gaussians



Strand-based Hairstyle

Unstructured Gaussians

Structured Gaussians



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“Gaussian Haircut ✂️”, ECCV 2024

Presented by Da

Inverse Rendering for Humans

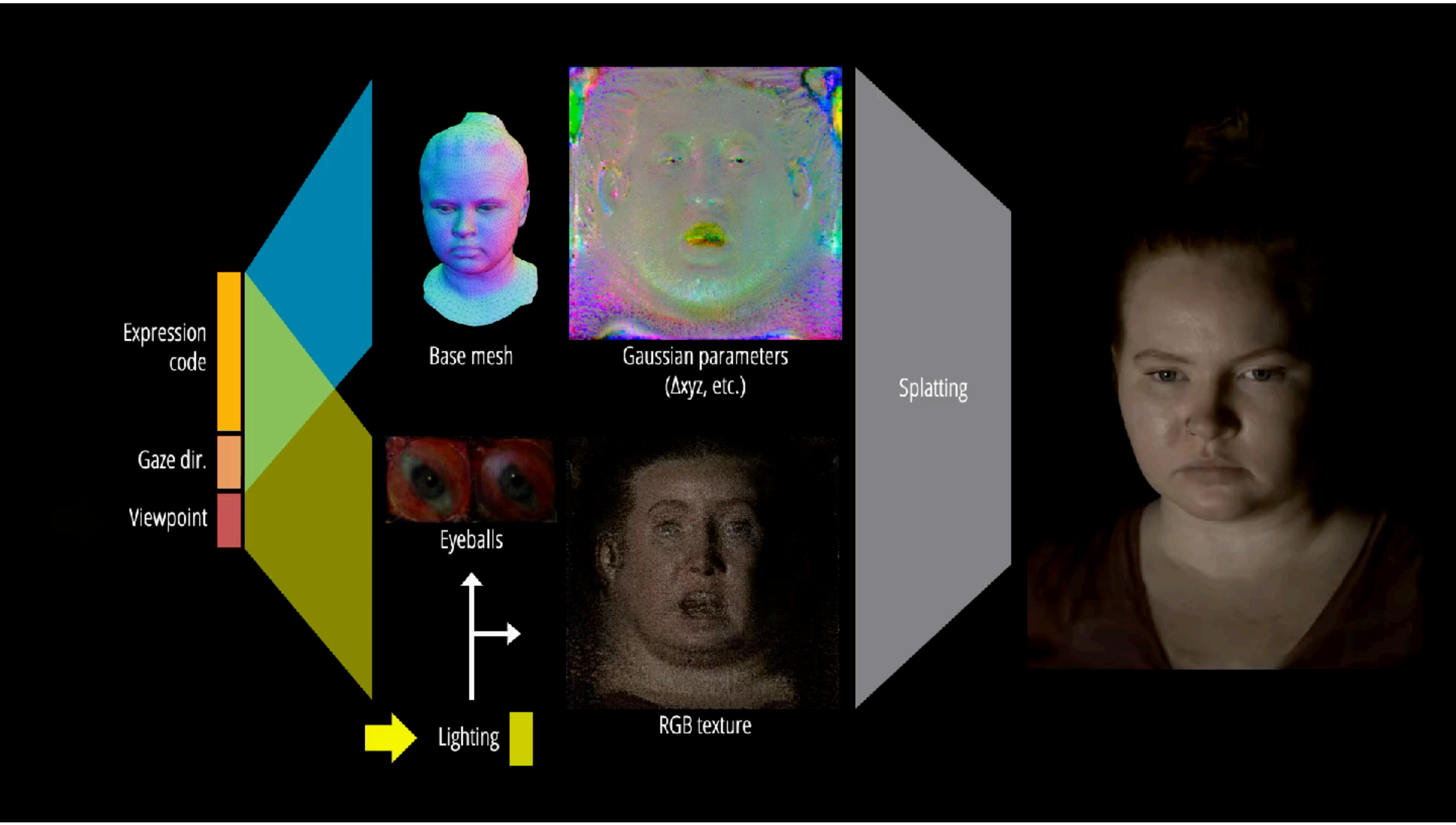
CS184

“Relightable Gaussian Codec Avatars”, CVPR 2024

Meta

Presented by Da

Inverse Rendering for Humans

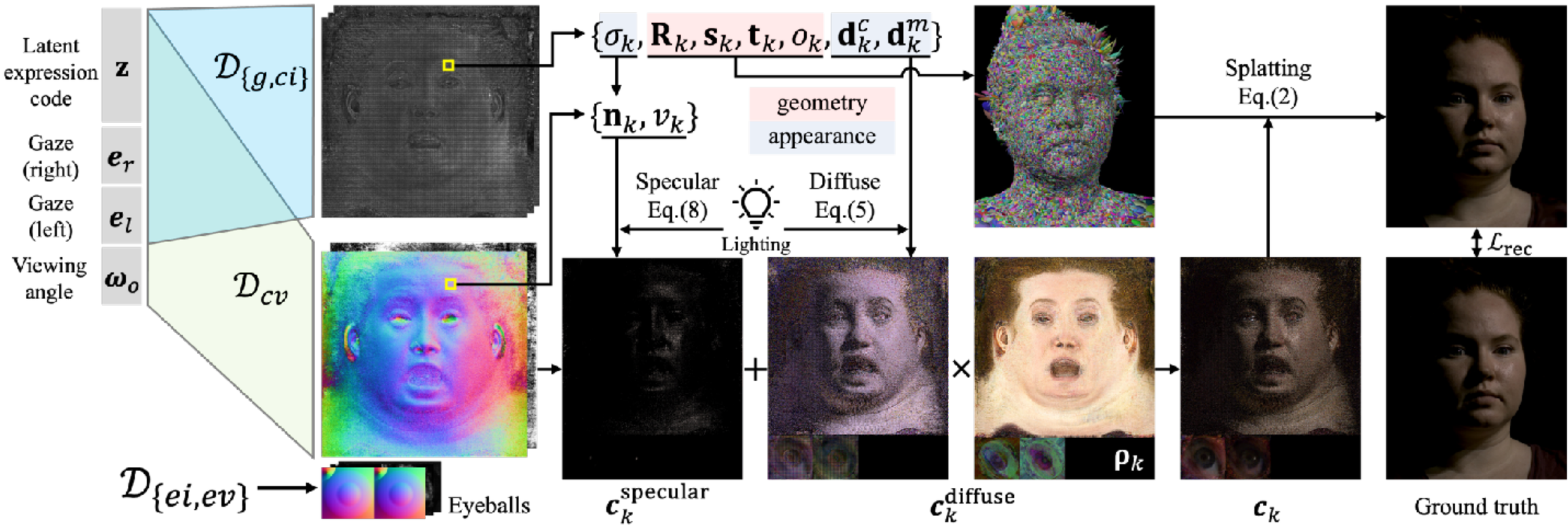


"Relightable Gaussian Codec Avatars", CVPR 2024

Meta

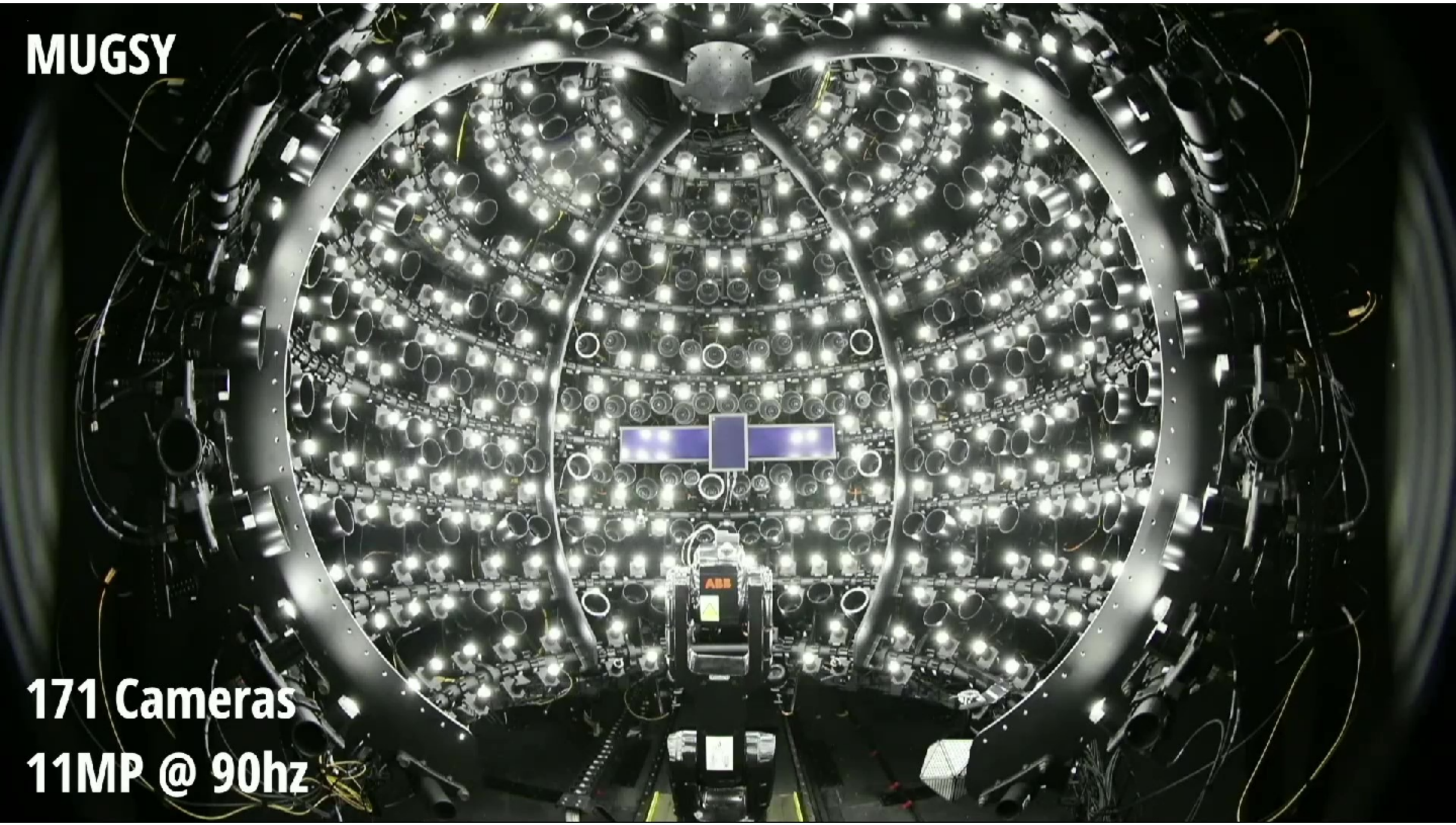
Presented by Da

Inverse Rendering for Humans



"Relightable Gaussian Codec Avatars", CVPR 2024

Inverse Rendering for Humans



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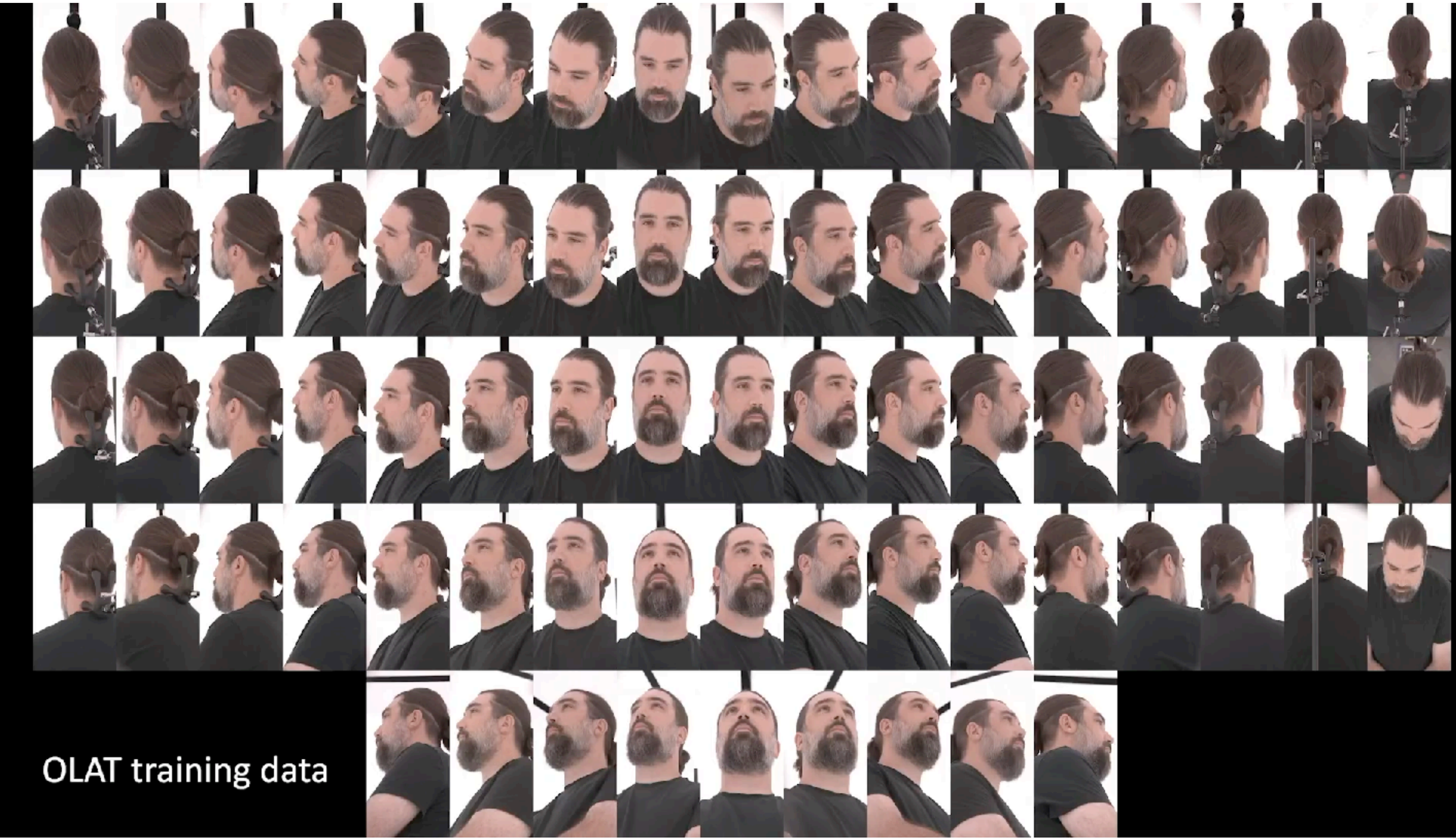
https://www.uploadvr.com/meta-codec-avatars-iphone-scan/

Meta

Presented by Da

Inverse Rendering for Humans

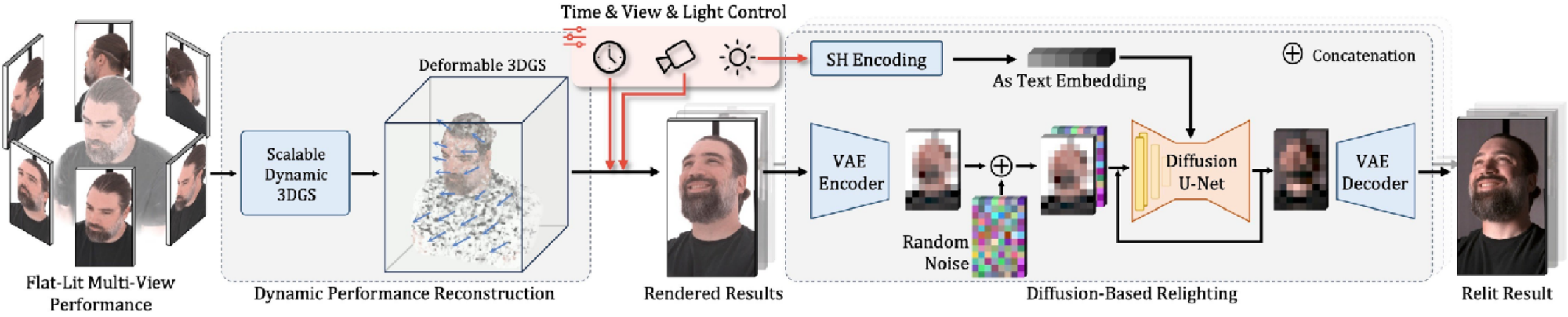


CS184

"DifFRelight", SIGGRAPH Asia 2024

EYELINE STUDIOS™
POWERED BY NETFLIX

Inverse Rendering for Humans



"DifFRelight", SIGGRAPH Asia 2024

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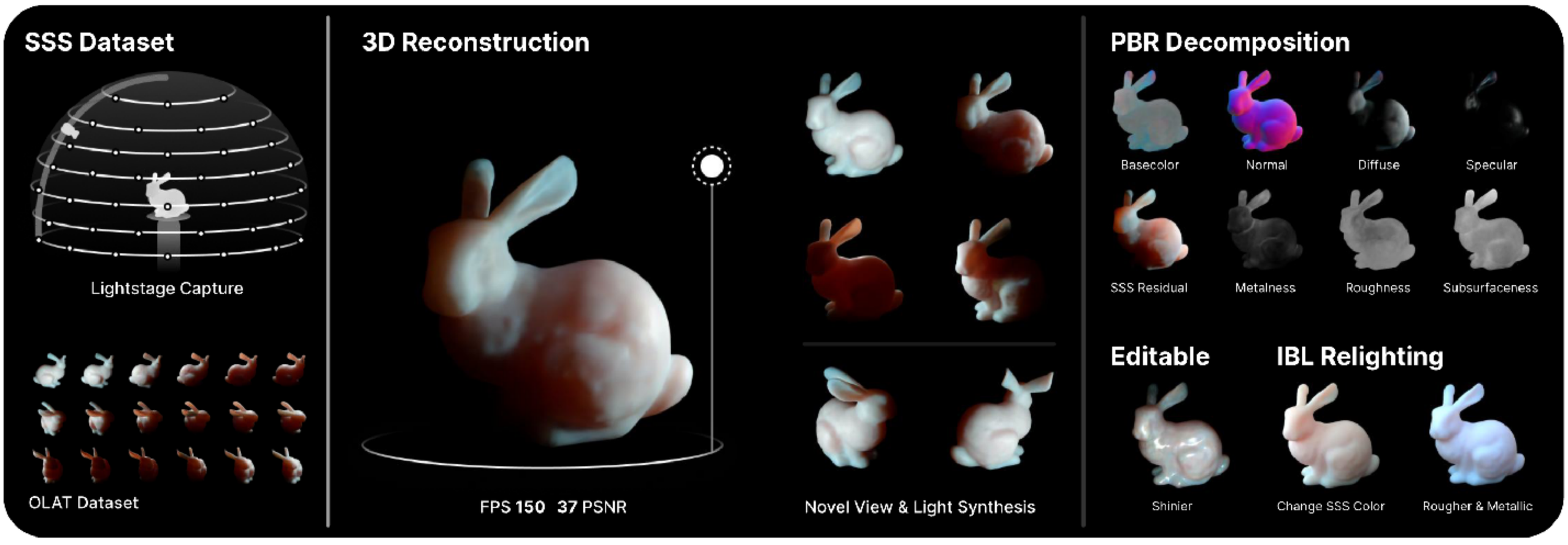
Inverse Rendering for Objects



CS184

"Subsurface Scattering for Gaussian Splatting" Presented by Da

Inverse Rendering for Objects

CS184

"Subsurface Scattering for Gaussian Splatting" Presented by Da

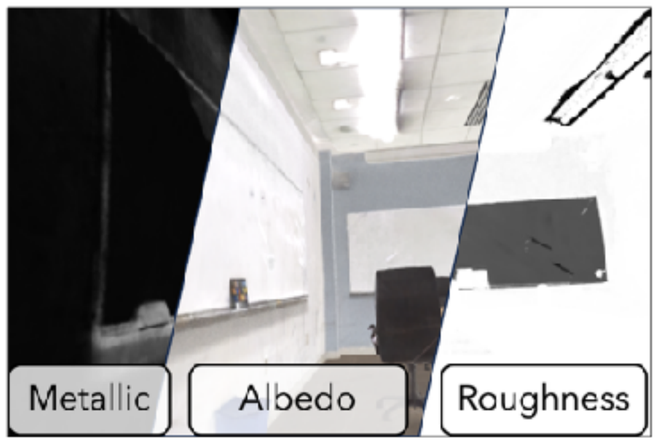
Inverse Rendering for Scenes

Input LDR Images

IRIS

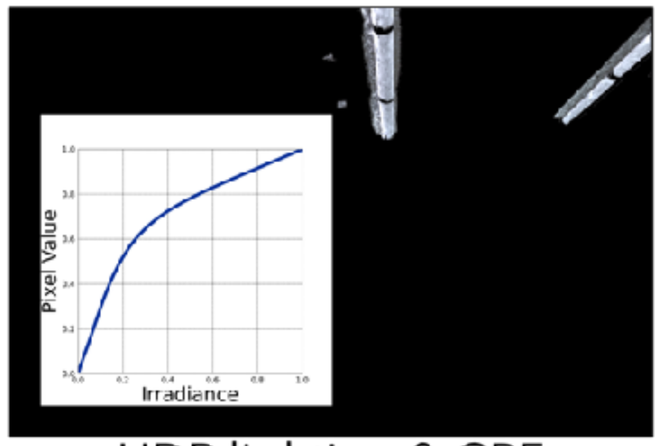
Geometry

Relighting

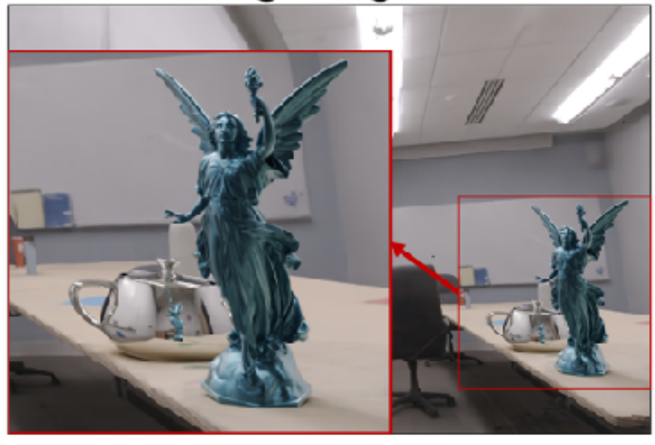


Surface material

Relighting



HDR lighting & CRF



Object Insertion

CS184

"IRIS: Inverse Rendering of Indoor Scenes" Presented by Da

Inverse Rendering for Scenes

Applications

- Relighting 1
- Relighting 2
- Object Insertion
- Original Lighting

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"IRIS: Inverse Rendering of Indoor Scenes" Presented by Da

Inverse Rendering for Scenes

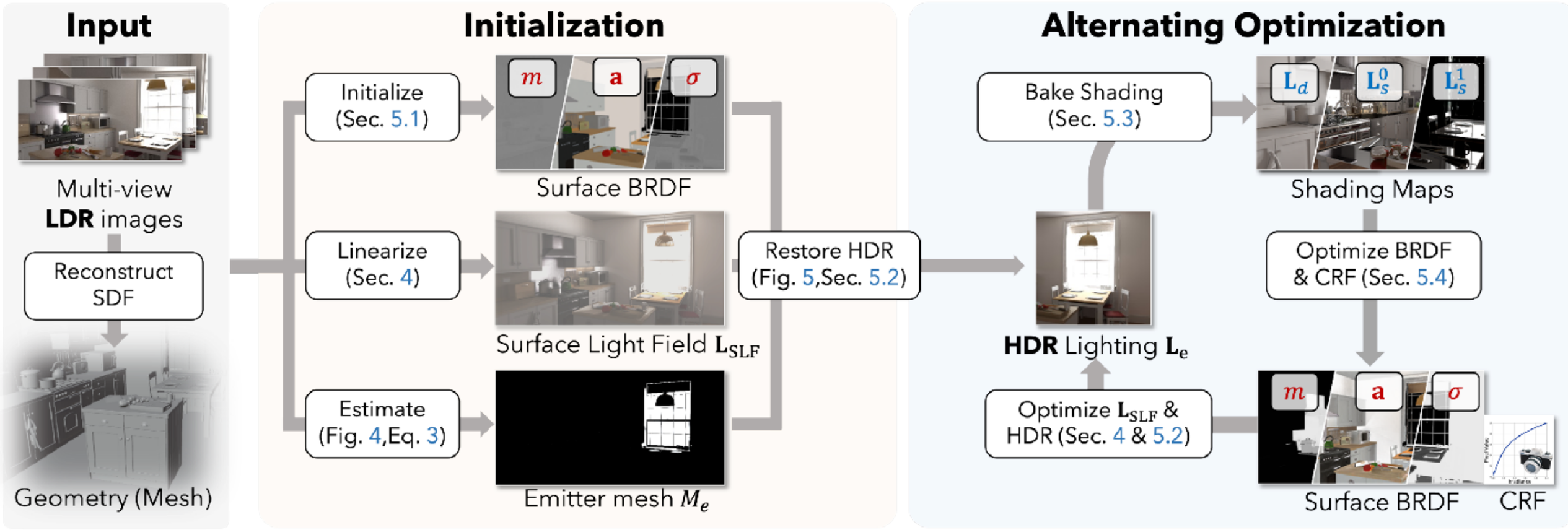
Results

- Diffuse Reflectance $\mathbf{k}_d$
- Material Reflectance $\mathbf{a}'$
- Roughness σ
- Metallic m
- Tonemapped HDR Emission $\mathbf{L}_e$
- Rerendering $\mathbf{L}$

CS184

"IRIS: Inverse Rendering of Indoor Scenes" **Presented by Da**

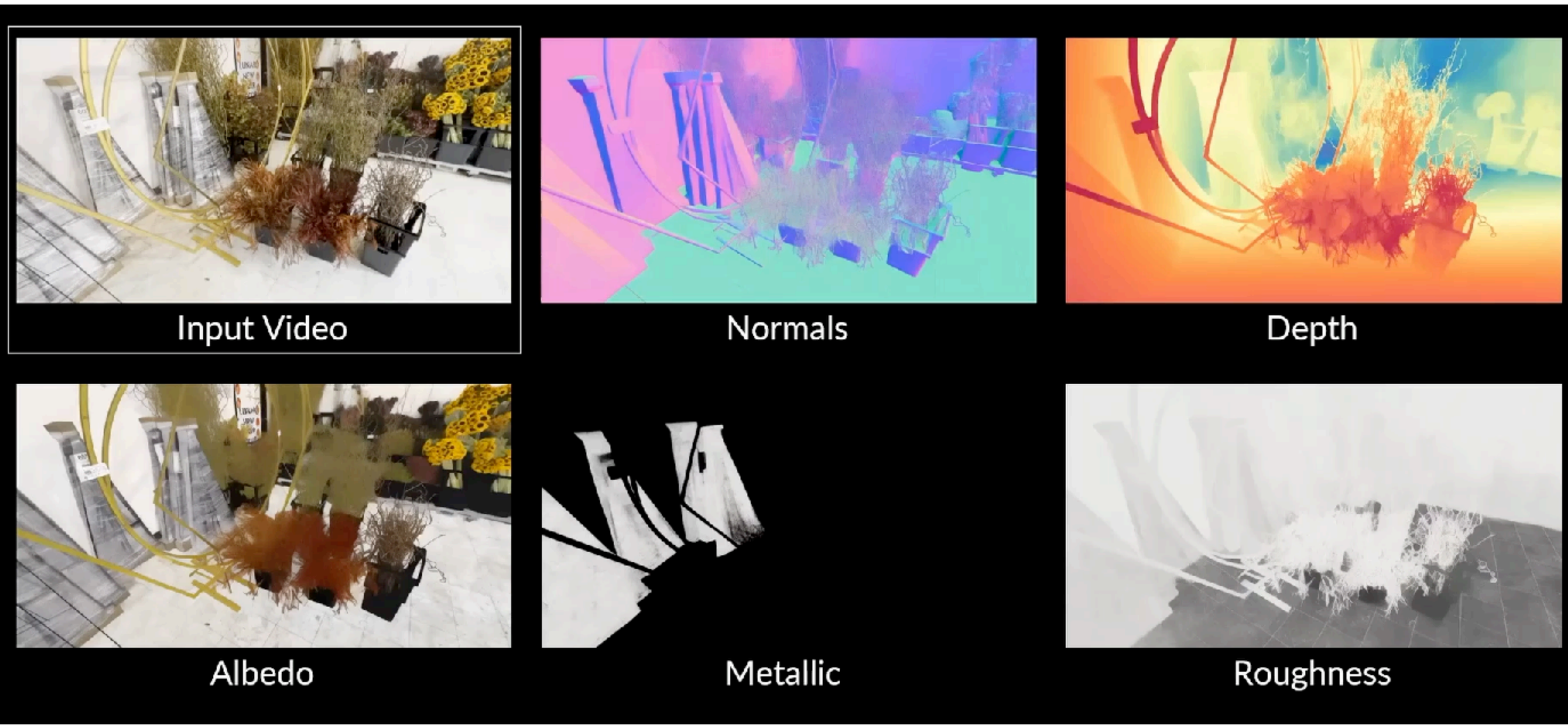
Inverse Rendering for Scenes



CS184

"IRIS: Inverse Rendering of Indoor Scenes" **Presented by Da**

Inverse Rendering for Scenes



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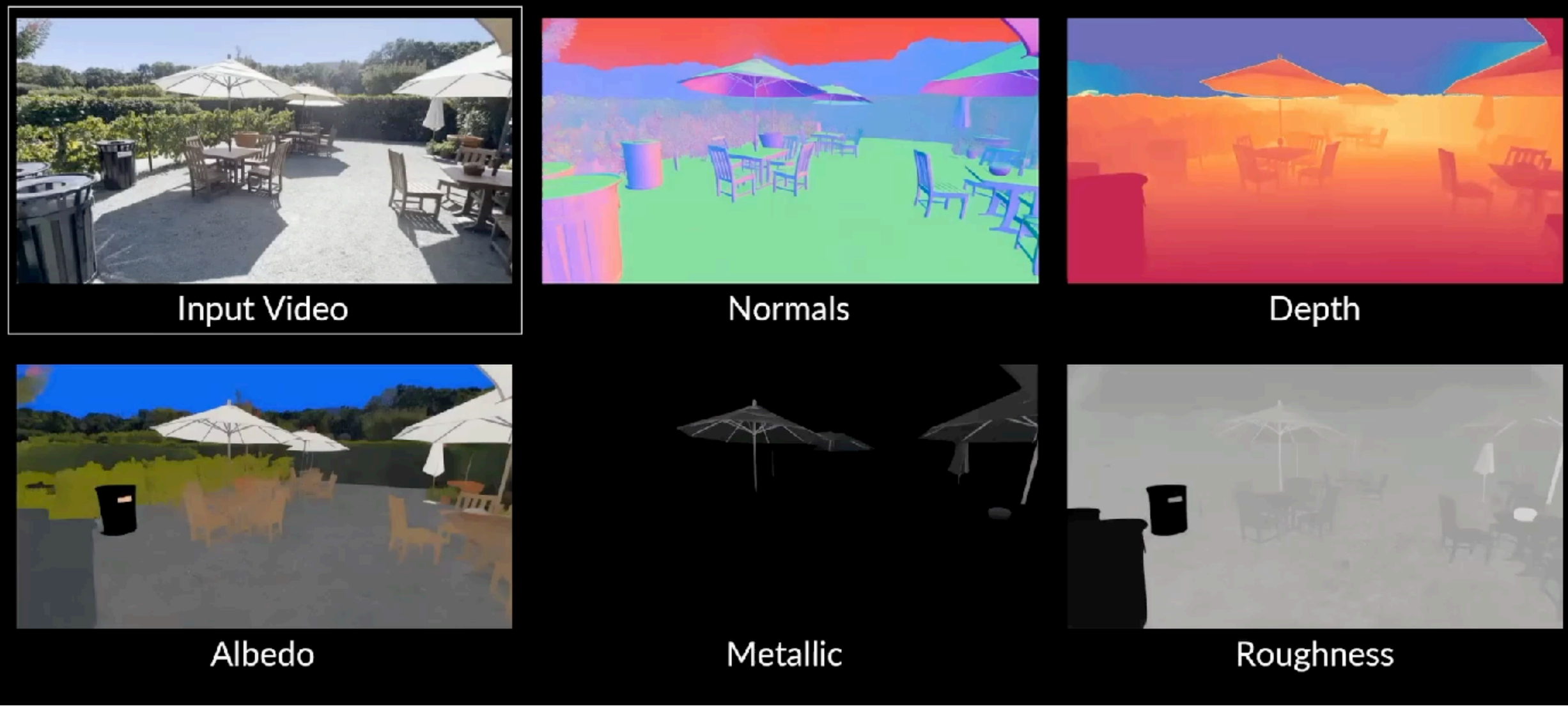
"DiffusionRenderer", CVPR 2025

Presented by Da

Inverse Rendering for Scenes



CS184

"DiffusionRenderer", CVPR 2025

Presented by Da

Inverse Rendering for Scenes

Object Insertion

Hover over each image to see the background without the virtual object.

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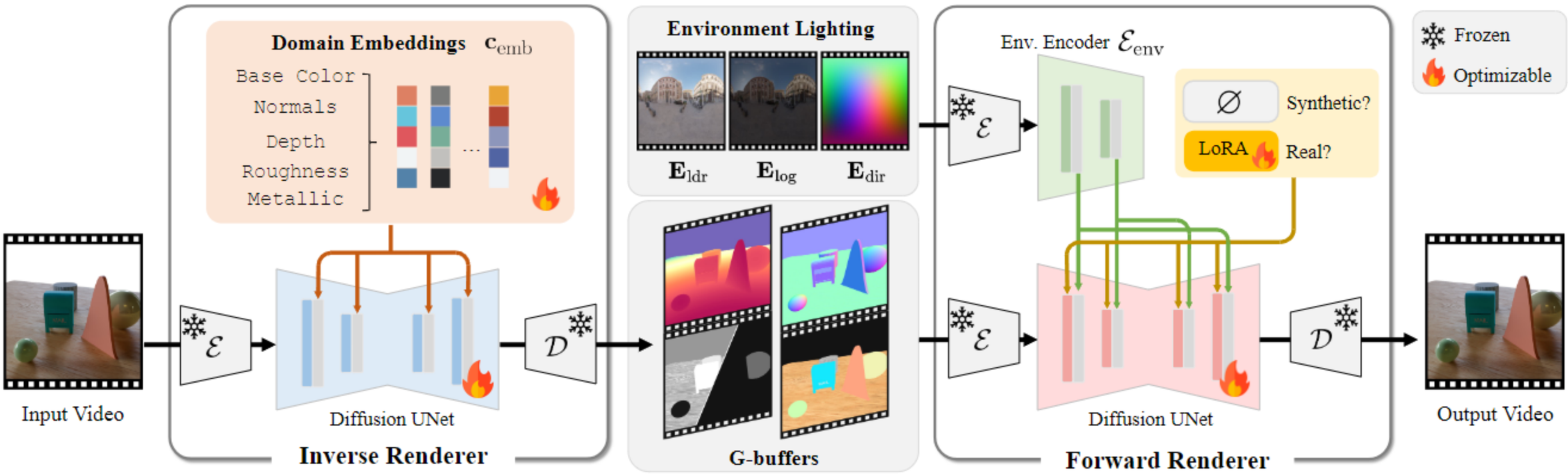
"DiffusionRenderer", CVPR 2025

Presented by Da

Inverse Rendering for Scenes



CS184

"DiffusionRenderer", CVPR 2025

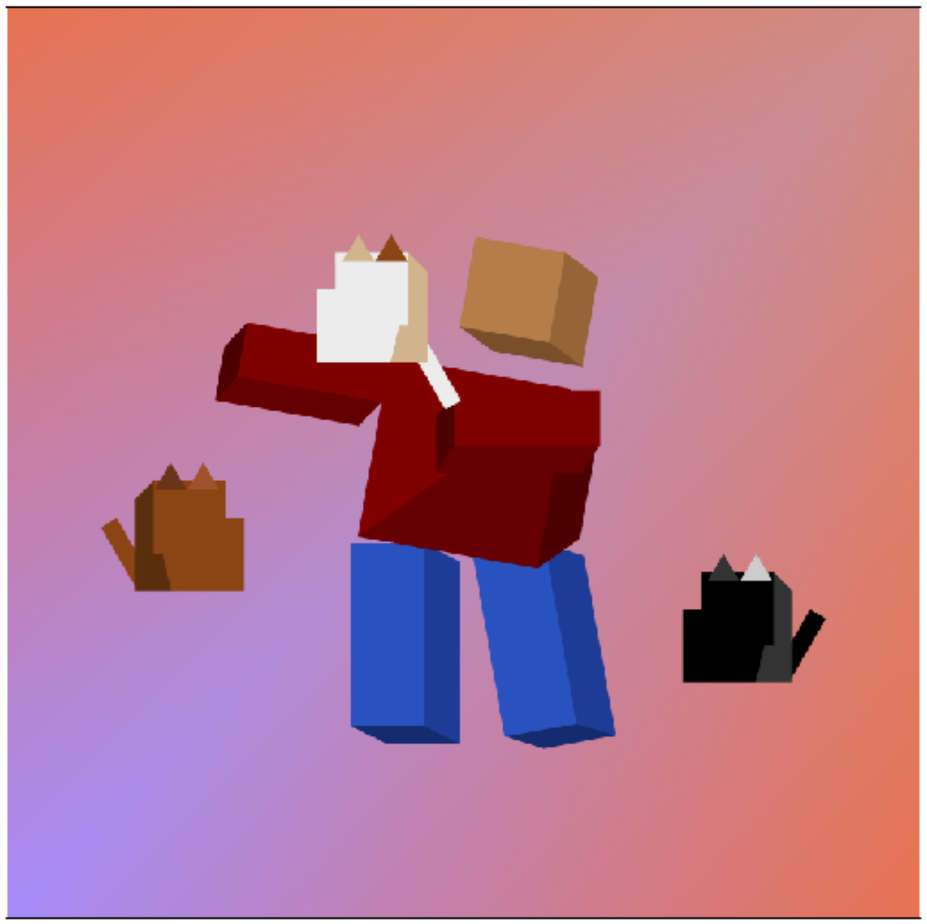
Presented by Da

Art Competition #1 Results

Art Competition #1 - 3rd Place Winner

Joel Castro

Catsterized



It started off as my robot for task 3. I got the idea to give it the illusion of depth by adding additional polygons and shading (filling) them appropriately. When I got done with that it occurred to me that there was a severe lack of cats so I applied to same techniques to block one out. I duplicated/mirrored/translated said blocked cat for a total of 3 distinct cats (after my 3 IRL cats). How did I know what coordinates to use throughout all of this? I used a special technique known to few called "spending 6 hours guessing and checking". Finally, I used colortri and two triangles to interpolate over a color pallet fitting the scene.

Art Competition #1 - 2nd Place Winner

Carolyn Liu & Lance Mathias

roar

inspo: https://drive.google.com/file/d/1unjTNuCE-i1MbrurwfqoiU3FVQFRXbZy/view?usp=sharing

1. drew 20 frames for my gif
2. converted the pngs to svgs using python image library
3. rendered images with the rasterizer created from hw1
4. used imagemagick to make a gif from the 20 screenshots

Art Competition #1 - 1st Place Winner

Shiran Yuan

Life is beautiful:)

1. Construct a canny map of the image. This is a transformation often used in computer vision that detects regions with high-frequency color changes.
2. Sample 1000 "edge points" from nonzero pixels of the canny map, and sample 100 random points across the entire image. Those two sets combined form the vertices.
3. Then perform Delaunay triangulation on the set of vertices.